Who Gains From Creative Destruction? Evidence From High-Quality Entrepreneurship in the United States

by

Astrid Marinoni
Georgia Institute of Technology

John Voorheis
U.S. Census Bureau

This paper is a revised version of "Who Gains from Creative Destruction? Evidence from High-Quality Entrepreneurship in the United States" (CES 19-29) from 2019 October. A copy of the original paper is available upon request.

CES 19-29R July 2026

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not necessarily represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

Papers in the series are posted on www.census.gov/library/working-papers/series/ces-wp.html. For information about the series, contact Sean Wang, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies, 4600 Silver Hill Road, Washington, DC 20233, CES.Working.Papers@census.gov.

Abstract

The question of who gains from high-quality entrepreneurship is crucial to understanding whether investments in innovative startup firms produce socially beneficial outcomes. Although entrepreneurship is widely viewed as a central driver of economic growth, recent research has raised concerns that it may also exacerbate inequality and other undesirable social outcomes. We bring new evidence to this debate by analyzing the impact of entrepreneurship on a variety of local economic outcomes, including income inequality, average incomes, and income mobility, using IRS and U.S. Census Bureau microdata. We find that entrepreneurship indeed increases income inequality, although this effect fades over time. While entrepreneurship produces statistically significant income gains across the distribution, the economic magnitude of these effects varies dramatically, with benefits concentrated among top earners. We also find that entrepreneurship serves as an important mobility tool for less well-off individuals, positively increasing their probability of rising to the top of the income distribution. These mobility effects are strong for Asian, Hispanic, and foreign-born individuals, but are less evident among Black individuals.

Keyword*:* entrepreneurship, innovation, income inequality

JEL Classification: L26, D63

---

* This paper is released to inform interested parties of research and to encourage discussion. Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7511151: CBDRB-FY18-164, CBDRB-FY19-342, CBDRB-FY19-509, CBDRB-FY20-348, CBDRB-FY20-389, and CBDRB-FY22-CES014-022.)

Marinoni: Georgia Institute of Technology, Scheller College of Business, Atlanta, GA; astrid.marinoni@scheller.gatech.ed, Voorheis: U.S. Census Bureau, Suitland, MD; john.l.voorheis@census.gov

1 Introduction

Entrepreneurial activity is central to economic growth (Schumpeter, 1942). As a result, policymakers have frequently introduced incentives to stimulate entrepreneurship, aiming to enhance economic development within their jurisdictions. In the past decade, millions of dollars have been spent on initiatives meant to nurture local entrepreneurs, with the aspiration of replicating the success of Silicon Valley, the most renowned entrepreneurial hub in the United States (Saxenian, 1994). At the same time, some recent studies have scrutinized the broader societal effects of entrepreneurship and highlighted some significant limitations (Brandtner et al., 2025), suggesting that the benefits of entrepreneurship are not as universally positive as previously assumed. In particular, several scholars hinted at the fact that entrepreneurship might be correlated with greater income inequality (Sorensen & Sorenson, 2007; Kwon & Sorenson, 2021), and might fail to create a significant number of jobs in the labor market (Fairlie et al., 2023).

The evidence linking entrepreneurship to inequality, however, leaves several important questions open. First, inequality alone is an incomplete guide to entrepreneurship's broader impact. If entrepreneurship raises incomes across the distribution, rising inequality is less detrimental than if the gains go only to the top. Assessing its impact therefore requires examining income *levels* across the distribution, as well as whether entrepreneurship reshuffles individuals' positions over time through income mobility. Second, this work typically measures entrepreneurship by its *quantity*, the number of new firms, a count that conflates high-growth ventures with subsistence and self-employment activity (Schoar, 2010), whereas the ventures that policymakers seek to cultivate are a small subset of high-quality, high-growth firms whose distributional footprint may differ from that of new businesses in general.[1] Third, the data required to study these questions, namely income distributions free of top-income censoring, have until recently been scarce. As a result, relatively little is known about who benefits from entrepreneurship across the income distribution, over time, and across segments of the population.

We address these gaps by examining who gains from high-quality entrepreneurship. To do so, we use a measure of entrepreneurship that captures venture *quality* rather than mere quantity (Andrews et al., 2022), together with new measures of income inequality and incomes built from U.S. tax microdata that are not subject to top-income censoring. With these data in hand, we trace the effect of high-quality entrepreneurship through the whole income distribution and over time, making four

[1] For instance, see OECD (2010), *High-Growth Enterprises: What Governments Can Do to Make a Difference*, OECD Studies on SMEs and Entrepreneurship, OECD Publishing, Paris, https://www.oecd.org/en/publications/high-growth-enterprises_9789264048782-en.html.

contributions. First, we examine its effect on income *inequality* and income shares; second, on *average incomes* at each point of the distribution, asking not only whether inequality rises but where in the distribution the gains fall and how large they are in dollar terms; third, on income *mobility*, asking whether entrepreneurship allows individuals to climb the income ladder or instead entrenches those already at the top; and fourth, on how these effects vary across *demographic groups*, by age, country of birth, and race.

In our analysis, we introduce and utilize new data sources. To gauge high-quality entrepreneurship, we use a measure that captures both entrepreneurial quality and entrepreneurial quantity (Andrews et al., 2022), in contrast to traditional approaches that use the number of businesses to measure entrepreneurship. This measure allows us to differentiate among ventures based on their probability of success, providing a guideline to distinguish entrepreneurs whose venture is a mere substitute for employment income from entrepreneurs whose goal is to generate high economic returns through an IPO or an acquisition. In addition, we construct new income and inequality measures at the commuting-zone (CZ) level, clusters of counties tied together by commuting flows that approximate local labor markets, from the IRS Form 1040 microdata, accessed through the U.S. Census Bureau, over the 1998–2014 period. Data on income distributions below the state level have been scarce or nonexistent, making studies of county- or CZ-level inequality difficult and forcing scholars to rely on approximation techniques (e.g., Aghion et al., 2019). Because they are built from tax microdata, these measures do not suffer from top-income censoring and let us examine the whole income distribution. We pair them with new state-level data that capture income mobility (Kondo et al., 2023).

With these data sources in hand, our baseline approach to estimating the effect of entrepreneurship on income inequality and income levels relies on a standard two-way fixed-effects specification that includes CZ-specific time trends, which absorb any shocks or changes that might happen at the CZ-level in a given year. It is possible, however, that the endogenous sorting of individuals across CZs over time might bias these estimates. To reduce this concern, we leverage a shift-share instrument for entrepreneurship by interacting a cross-sectional measure of industry shares with national-level trends in entrepreneurship (Bartik, 1991). We show that this variable passes all the recent robustness checks dedicated to this class of instruments (Adao et al., 2019; Borusyak et al., 2020; Goldsmith-Pinkham et al., 2020).

The first contribution of the paper is to show that, in line with previous work (e.g., Sorensen & Sorenson, 2007) that considers entrepreneurship more broadly, even high-growth entrepreneurship has a positive and statistically significant effect on *income inequality*. Our main IV specification suggests that a 1-point increase in entrepreneurship, which roughly corresponds to the establishment

of 10 high-quality businesses[2] is associated with a 0.0013 increase in the Gini coefficient after five years. We further examine the dynamics behind this result by estimating the relationship between entrepreneurship and ordinates of the Lorenz curve, or equivalently, top income shares. These results indicate that this relationship is concentrated at the top of the distribution, with high-quality entrepreneurship associated with a larger share of income going to individuals above its midpoint. Despite this initial effect, we find that the upward pressure on inequality from entrepreneurship fades over time.

A second key insight of the paper is that entrepreneurship is associated with rising *incomes* across the distribution, even though these effects are economically meaningful only for those already at the top. Instrumented specifications suggest that a 1-point increase in entrepreneurship is linked with a 7 percent higher average income for individuals belonging to the first decile after five years. This figure declines to 1 percent when considering the income of individuals at the median of the distribution and rises again to 2.5 percent when taking into account individuals belonging to the top 1 percent. However, translating these percentages into dollar values reveals a stark concentration of economic gains: the first decile gains $40 over five years, the fifth decile gains $288, while the top 1 percent gains ≈$26,300, a ratio of 650:1 between top and bottom. This pattern reconciles seemingly contradictory findings in the literature: entrepreneurship is associated with broadly higher incomes in relative terms (e.g., Schumpeter, 1942; Glaeser et al., 2015), but also with greater inequality, because gains in the middle of the distribution lag behind the income-weighted average while the bottom's share of total income remains small, so that absolute dollar gains are overwhelmingly captured by those already advantaged.

Taking our analysis a step further, we show that these positive associations are specific to high-quality entrepreneurship, while merely considering the number of new incorporations shows a much weaker relationship. We also find that these effects are distributed across several industries rather than being concentrated in a few categories. However, the income of individuals at the left tail of the income distribution rises the most when considering industries employing lower-skilled workers. When examining the impact of entrepreneurship on average incomes by demographic characteristics, we find that individuals aged 25 to 44 benefit the most. In terms of race, entrepreneurship is associated with the largest income gains for low-income American Indian and Alaska Natives (AIAN) and Asian individuals, while for Black and African American individuals the positive association appears mostly on the right section of the income distribution. Lastly, when considering the country of birth, foreign-

[2]We define a high-quality startup as an incorporated company in a high-tech sector, with a patent, a trademark, and incorporated in Delaware (Guzman & Stern, 2020).

born individuals tend to benefit more than natives, although this advantage diminishes at both the very low and very high ends of the income distribution.

As a third contribution, we move beyond income levels to explore how entrepreneurship affects *income mobility* (Carroll & Chen, 2016). This analysis is particularly informative as it reveals whether the observed increase in inequality is due to reranking (i.e., individuals moving from lower to higher income brackets) or whether it is simply due to high-income individuals further improving their already favorable positions. Our findings indicate that individuals initially in the 1st to 25th percentile have a significant and positive probability of reaching the 75th percentile or higher after five years, suggesting that entrepreneurship can act as a tool for upward mobility among individuals in the lower income ranges. This upward mobility effect is however counterbalanced by individuals who start in the 50th to 75th percentile range, who experience a positive probability of falling to the 1st to 25th percentile range after five years.

As a fourth contribution, we examine heterogeneity in income mobility across demographic groups, thereby extending research on how different populations engage with and benefit from entrepreneurship. Prior studies have shown that minority entrepreneurs face substantial structural barriers, including limited access to capital, discriminatory treatment, and weaker networks (Cook et al., 2022), that affect both entry into entrepreneurship and long-term success. These constraints have been found to impact groups unevenly: while some, such as Asians, tend to thrive in entrepreneurial settings, others, such as African Americans, often encounter more persistent challenges (Fairlie & Robb, 2010). Our results provide new evidence on these patterns by focusing on income mobility and by distinguishing whether mobility gains stem from entrepreneurial income or wage earnings. We find that entrepreneurship serves as a strong income ladder for individuals aged 25 to 44 and for Asian, Hispanic, and White individuals. For Asians and Hispanics, mobility is primarily driven by wage income rather than income coming directly from business ownership or investments. American Indian and Alaska Native (AIAN) and Black and African American individuals generally experience negative mobility effects: while entrepreneurship can improve the financial situation of these groups, it does not provide the substantial economic push needed for significant upward mobility compared to others. Going deeper into the reasons behind this result, we find that the economic mobility of Black and African American individuals is primarily hindered by wage-related dynamics rather than business ownership, highlighting the existence of significant wage gaps between groups (Chetty et al., 2020), perhaps driven by network-related distortions in hiring (Ruef, 2010). Regarding country of birth, foreign-born individuals, who have been found to play an important role in high-growth startups both as employees and entrepreneurs (Kerr & Kerr, 2020),

exhibit a much higher probability of upward income mobility than natives. Taken together, these findings deepen our understanding of who benefits from entrepreneurship and through which economic channels, offering a more comprehensive view of the role of entrepreneurship in promoting inclusion and mobility.

2 Entrepreneurship and Incomes

Entrepreneurship operates within existing economic structures that shape who gains access to opportunities and how economic rewards are distributed. Those with greater financial resources, professional networks, and access to capital markets are better positioned to identify opportunities and mobilize resources, while others face substantial barriers to entry and growth (Bell et al., 2019). These dynamics mean that entrepreneurship actively participates in structuring how opportunity and income are allocated across the economy.

Understanding these distributional consequences matters because inequality itself shapes broader economic trajectories. Societies marked by stark income disparities experience lower long-term economic growth (e.g., Alesina & Rodrik, 1994; Galor & Moav, 2004) and reduced social mobility (e.g., Corak, 2013). Whether entrepreneurship contributes to shared prosperity or instead reinforces existing divides depends on the mechanisms through which it diffuses benefits, and shapes prospects for economic mobility.

Concentration of returns. A first mechanism through which entrepreneurship influences inequality is by concentrating economic rewards among a small subset of actors. High-quality, high-risk ventures yield returns that far exceed market averages, particularly in high-tech and STEM-intensive sectors where entrepreneurs often experience substantial income premiums (Åstebro et al., 2011; Braguinsky et al., 2012). Standard economic theory predicts that activities involving heightened risk or uncertainty must yield higher compensation to attract participation. Because these returns diverge sharply from the population mean, they widen the income distribution and amplify inequality.

These dynamics are further intensified by structural features of entrepreneurial markets. Entrepreneurship is characterized by heavy-tailed outcome distributions: while most new ventures fail or remain small, a small fraction succeed spectacularly, generating extraordinary returns. This winner-take-all logic, in which outcomes are highly skewed and a minority of firms capture a disproportionate share of rewards, is particularly pronounced in industries characterized by scalability and network effects, such as digital technologies (Frank & Cook, 1996). Early advantages compound as access to capital, talent, and market visibility accelerates growth, reinforcing the position of leading firms (Lieberman & Montgomery, 1988). Venture capital activity, which

disproportionately targets high-growth ventures, further amplifies these cumulative advantages and is strongly associated with rising wage inequality across regions (Kwon & Sorenson, 2021). Together, these self-reinforcing feedback loops mean that entrepreneurial markets tend to concentrate economic rewards among a relatively small number of actors.

Labor market restructuring and diffusion. A second mechanism operates differently: entrepreneurship can diffuse economic benefits beyond founders and investors (including employees, suppliers, and other stakeholders in the entrepreneurial ecosystem) and contribute to rising incomes across the distribution. Entrepreneurial activity has long been identified as a driver of technological progress, productivity improvements, and broader economic expansion (e.g., Romer, 1990). New and growing firms play an outsized role in employment creation, accounting for a substantial share of gross firm-level job growth (Neumark et al., 2011; Haltiwanger et al., 2013; Decker et al., 2014). As these firms emerge and scale, they reshape local employment structures by generating new positions and expanding labor demand across a wide range of occupations (Glaeser et al., 2015). Business creation also generates benefits related to agglomeration economies, as high entrepreneurship is associated with thicker labor markets, knowledge spillovers, and increases in overall productivity (Marshall, 1920; Delgado et al., 2010).

These dynamics extend beyond the number of jobs created to the quality and distribution of those jobs. Rapidly scaling firms demand a diverse mix of technical, managerial, and operational skills, generating both high-wage professional opportunities and complementary roles that broaden access to productive employment (Autor & Handel, 2013). Firm-level characteristics such as size, workforce composition, and rates of growth shape wage-setting within organizations, with implications that scale up to broader income distributions (e.g., Sorensen & Sorenson, 2007). Moreover, compensation dispersion within firms and inequality in access to social capital can influence employee mobility and entrepreneurial exits, as these forms of inequality may motivate individuals to seek opportunities elsewhere or start their own ventures (Carnahan et al., 2012; Sorensen & Sharkey, 2014). Empirical evidence on wage premiums in startups is mixed: while some studies find that younger workers can earn more in innovative startups relative to similarly qualified workers in older firms (Ouimet & Zarutskie, 2014), and that some rapidly growing young firms offer wage premiums even after accounting for worker characteristics (Babina et al., 2019; Schmieder, 2023), other research using within-individual comparisons suggests that apparent wage premiums may largely reflect worker selection rather than firm effects (Kim, 2018).

High-growth entrepreneurship can also transform labor markets by increasing the value of work. As startups introduce new technologies and production methods, they raise local productivity levels,

which in turn enhances firms' willingness and ability to pay higher wages (Van Reenen, 1996; Acemoglu, 2002). High-growth ventures compete intensely for talent, offering compensation packages that often exceed those available in incumbent firms. This competition pushes existing employers to revise compensation upward, elevating pay standards even for workers not directly employed by new ventures (Combes et al., 2010). The clustering of innovative firms amplifies these dynamics by increasing labor market thickness, strengthening demand for skilled labor, and improving workers' bargaining power and mobility (Moretti, 2012). Recent evidence documents increasing wage dispersion between more and less productive firms (Berlingieri et al., 2024), suggesting that entrepreneurship may widen income distributions by amplifying differences between frontier and lagging establishments.

Importantly, benefits may not be confined to high-skilled workers. The formation and expansion of new firms stimulate demand in complementary industries such as logistics, supply, professional support, retail, and maintenance, generating employment opportunities for workers with diverse skill levels (Moretti, 2010; Eeckhout et al., 2014). High-growth entrepreneurial activity may therefore raise wages even for lower-income workers by expanding local demand and generating spillovers into adjacent sectors (Glaeser et al., 2015). Through these labor market transformations, entrepreneurship can diffuse economic benefits and elevate incomes across the distribution, offering a countervailing mechanism to its inequality-increasing effects through returns to ownership.

Mobility dynamics. Beyond static distributional outcomes, the impact of entrepreneurship on economic mobility deserves attention. Schumpeter (1942) highlighted entrepreneurship's potential to disrupt established positions and open new avenues for advancement through creative destruction. New firm formation can act as a vehicle for mobility if it enables individuals from lower-income backgrounds to improve their economic standing, whether through ownership or through employment in rapidly growing firms.

This potential, however, depends critically on who participates and how. If entrepreneurial entry is concentrated among those with existing wealth, networks, and knowledge, new ventures may simply channel further gains to the already advantaged, reinforcing existing hierarchies. Moreover, entrepreneurship's impact on mobility is unlikely to be uniform. Some individuals and groups may move upward, while others stagnate or fall, resulting in a reordering of positions rather than a reduction of inequality. Economic mobility can occur through both ownership pathways and wage employment. High-growth startups often require diverse skills and positions, which can lead to substantial wage growth for employees at various levels of the income distribution (Eeckhout et al., 2014; Lee, 2017). The mobility effects of entrepreneurship therefore depend on access to both entrepreneurial opportunities and to employment in high-growth firms.

Structural barriers and demographic heterogeneity. The above considerations assume that individuals have equal access to opportunities generated by entrepreneurship. In reality, deeply embedded disparities in access to financial capital, education, networks, and exposure to innovation shape both who becomes an entrepreneur and who is positioned to benefit from entrepreneurial employment. Prior research shows that individuals from more advantaged socioeconomic backgrounds are far more likely to enter innovative occupations, even when underlying abilities are similar (Bell et al., 2019).

These unequal starting points have important implications for who benefits from entrepreneurship. For Black and African American individuals, historical and contemporary barriers include restricted access to startup capital (Fairlie & Robb, 2010; Cook et al., 2022), weaker networks to investors and mentors, occupational segregation, and persistent wage gaps in employment (Bates, 1997; Cajner et al., 2017; Chetty et al., 2020). These constraints may limit both pathways through which entrepreneurship affects income: direct participation as founders or investors and indirect benefits through employment in high-growth ventures. Importantly, wage-related barriers may operate independently of entrepreneurial opportunities, meaning that even in regions with high entrepreneurial activity, some workers may experience smaller wage gains due to occupational sorting or labor market discrimination. For instance, Hispanic individuals may face language and documentation barriers affecting employment access in entrepreneurial sectors, while they are also overrepresented in certain entrepreneurial industries (Fairlie, 2018). Asian individuals, particularly in technical fields, often have strong representation in high-growth startups as employees (Kerr & Kerr, 2016) but may face barriers to entrepreneurial leadership (Lu et al., 2020). These distinct structural positions suggest that entrepreneurship's effects on income and mobility will vary not just in magnitude but in the channels through which benefits flow, with some groups benefiting primarily through employment, others through ownership, and still others facing constraints in both domains. Age also matters: younger workers are more likely to be employed in startups, positioning them to benefit from entrepreneurial growth (Ouimet & Zarutskie, 2014).

In summary, the relationship between entrepreneurship and income is complex and multifaceted. While entrepreneurship has the potential to drive economic growth, create jobs, and increase productivity, it can also concentrate economic gains among a small segment of the population. Understanding its full impact requires examining not only aggregate inequality but also income levels across the distribution and mobility patterns across demographic groups. Table A1 in the Appendix brings these threads together, mapping each of the four mechanisms above to the empirical analysis that tests it and to the corresponding finding.

3 Data

3.1 Entrepreneurial Quality

Our data on entrepreneurship are from Andrews et al. (2019). They measure entrepreneurial quality by linking the probability of a growth outcome as a function of startups' characteristics observable at or near the time of initial business registration.[3]

They construct the entrepreneurship dataset as follows. First, they estimate the relationship between startup features at founding and successful growth outcomes (an IPO or a significant acquisition). They use predictions from these estimates to generate an average quality score for each ZIP code in the United States for the 1988–2014 period. By multiplying this index by the number of businesses established in a given area-year combination, they obtain an overall score (the RECPI index).

This measure allows us to assess the potential of an entrepreneurial venture at the moment of founding, making it a valuable tool for identifying firms likely to generate both significant community impact and strong returns for their founders. While it is correlated with both venture capital activity and successful exits, it is not limited to capturing VC-backed firms, but encompasses a much wider spectrum of successful ventures. RECPI reflects quality at founding and thus includes high-quality startups that may never undergo an IPO or acquisition, not only those that eventually exit. Given its ability to capture a broad set of high-growth firms, the measure has been widely adopted in the literature (e.g., Guzman & Kacperczyk, 2019).[4]

In Section A-2 of the Appendix, we provide further analyses showing how this measure differs from the simple number of establishments and how it instead is more aligned with measures capturing VC-backed entrepreneurship, two checks that support its validity as a measure of high-growth entrepreneurship.

3.2 Income Inequality, Average Incomes, and Mobility

To proceed with our analysis, we require data on the income distribution within CZs coinciding with the period for which data on high-quality entrepreneurship are available. The commuting zone is our preferred unit because it approximates a local labor market: CZs are clusters of counties tied together by commuting flows, so that residents largely both live and work within the same zone (Tolbert & Sizer,

[3]For instance, they consider whether the firm is named after the entrepreneur, whether it is organized to facilitate equity financing (e.g., registering as a corporation), and whether the firm develops innovations or trademarks.

[4]The entrepreneurial index is available for 49 states. While the original data are available down to the ZIP code level, here we aggregate these data to the Commuting Zone (CZ) level, which is our preferred geographical unit of analysis. We use the CZ 2000 definition for CZs and we transform entrepreneurship from ZIP level to CZ level using David Dorn's crosswalk: https://www.ddorn.net/data.htm. Given that CZs sometimes span different states, we discard CZs that contain states for which one or more underlying entrepreneurial variables at the ZIP code level are not available.

1996). The local spillovers of high-quality entrepreneurship, such as new jobs, wage pressure, and local demand, operate at this scale, and the CZ is the standard geography in research on local labor markets and economic opportunity (Autor et al., 2013; Chetty et al., 2014; Chetty et al., 2020). Data at this level, however, are difficult to obtain: no suitable official estimates exist of income-related outcomes at sub-state-level geographies. To address this, we produce a set of new, small-area income and inequality measures using the universe of IRS Form 1040 tax returns. The Census Bureau is authorized by the IRS to use an extract of these files, containing information on tax-unit characteristics and geography as well as limited information on income received by tax units, to perform research that benefits Census operations and production activities. These files are available for the 1998–2014 period. It is important to note that in contrast to other income-related sources (e.g., the American Community Survey), the income concept (adjusted gross income (AGI), which is all pretax income minus above-the-line deductions) in the IRS Form 1040 tax returns *includes* realized capital gains, which may play a fundamental role in entrepreneurs' income.

To examine the relationship between entrepreneurship and inequality, we calculate several measures of income inequality at the CZ level. The most familiar of these is the well-known Gini coefficient, calculated as $G = \frac{1}{\mu_x 2n^2} \sum_{i=1}^{n} \sum_{i'=1}^{n} |x_i - x_{i'}|$. The Gini coefficient is a commonly used measure of overall income inequality, but it does not necessarily reveal which parts of the income distribution are most affected by a particular factor. One way to examine this is to directly use the ordinates of the Lorenz curve as measures of income inequality. The pth ordinate of a CZ's Lorenz curve is $L(p) = \frac{1}{\mu} \int_0^{F^{-1}(p)} x f(x)\, dx$, which we estimate using the Binder and Kovacevic (1995) estimator. These Lorenz curve ordinates can be interpreted as the share of income accruing to the bottom pth percent of the income distribution, and so $1 - L(p)$ can be interpreted as top income shares, à la Piketty and Saez (2003).[5]

To shed light on whether entrepreneurship is associated with higher incomes, we leverage the information on average incomes for different percentiles in the income distribution by using the universe of IRS Form 1040 tax returns. To delve deeper into our analysis and shed light on who gains the most from entrepreneurship, we will leverage income data similar to those above but broken down by demographic features.[6]

To capture income mobility, we leverage data from the Income Transition Matrix (Kondo et al., 2023). This dataset provides statistics on the probability that an individual starting in a given income

[5]Both the Gini coefficient and the Lorenz curve ordinates capture a relative inequality concept. For each inequality measure, we adjust for the size of the income-receiving unit using a square-root equivalence scale.
[6]Because of Census restrictions, these data are only available at the state level (Kondo et al., 2023).

bin[7] moves to another income bin after five years. These transition matrices reveal the likelihood that individuals from various demographic groups will move between income bins over time, offering a comprehensive view of economic mobility. Because mobility data are available only at the state level, these results should be interpreted with caution due to the aggregation of diverse regional economies within states. Because we vary the geographic unit of analysis across the paper, we interpret the mobility results below as associations rather than as causal estimates.

3.3 Descriptive Statistics

Of immediate interest is the naive correlation between income inequality, income levels, and high-growth entrepreneurship over time and space. Figure 1 reports two scatterplots mapping the entrepreneurial index and the Gini index (Panel A) as well as the average incomes of individuals at the median of the income distribution (Panel B) for selected years. A superficial visual inspection suggests a positive relationship in both cases. California plays a dominant role: the Los Angeles and San Francisco Commuting Zones display two of the highest values for both entrepreneurship as well as inequality and income. Boston is also responsible for driving the positive relationship in both graphs. These trends are quite consistent across the years and are in general robust to the exclusion of California. Overall, the relationship is positive and is not sensitive to outliers.

Figure A1 in the Appendix considers similar correlations at the spatial level for the year 2014. Once again, Panel A maps the bivariate relationship between entrepreneurship and Gini, while Panel B considers the average incomes of individuals at the median of the income distribution. Both figures consider different percentiles of both measures. Dark blue denotes those CZs with a substantially high value (in the top third of both distributions) of both variables of interest (i.e., entrepreneurship and inequality in Panel A, and entrepreneurship and income of individuals at the median in Panel B).

Not surprisingly, four areas are driving the relationship between entrepreneurship and inequality: California, part of the Eastern Seaboard, Texas, and Florida. Meanwhile, high inequality and low entrepreneurship are present throughout the Central states, while areas with high entrepreneurship and low inequality are scattered around the country. Panel B confirms the positive trends driven by California, Texas, and the Northeast to a lesser extent. At the same time, the graph shows how several CZs in the Midwest are exhibiting high values of income and medium to high levels of entrepreneurship. The Southeast instead consistently shows low and medium income levels despite several areas being high in entrepreneurship. Finally, Table 1 shows some basic descriptive statistics at the CZ level for the period of our analysis (1998–2014).

[7] The income bins (quartiles of the state-level income distribution) are defined across all members in a sample, year, and geography.

4 Empirical Strategy

Our goal is to identify the effect of CZ-level entrepreneurship, measured by RECPI (the entrepreneurial index), on the level and distribution of income within CZs, captured by the Gini index. Given that the effects of entrepreneurship on incomes, and specifically on income inequality, might occur a few years after the establishment of high-growth startups, we estimate a two-way fixed-effects regression with distributed lags (Dube, 2019) considering a five-year horizon.[8] In general, this model is akin to a standard fixed-effects model, but it includes lagged values of entrepreneurship as follows:

$$Y_{i,t} = \alpha + \sum_{k=0}^{n} \beta_k Entrepreneurship_{i,t-k} + X_{i,t}\Gamma + \lambda_i + \delta_t + \theta_{i,t} + e_{i,t} \quad (1)$$

where i indexes CZs and t indexes years. As the main dependent variable, $Y_{i,t}$, we will consider both the Gini index as well as the average income of individuals at the median point of the income distribution (i.e., average income at the 5th decile of the income distribution). $Entrepreneurship_{i,t-k}$ is measured with the entrepreneurial index with a cumulative lag of 5 years. The effect of entrepreneurship on the dependent variable will then be equal to the sum of the contemporaneous up to five-year lagged RECPI coefficients.[9] $X_{i,t}$ is a set of controls at the CZ level that includes population, unemployment, race, age, GDP, female population, patents, GDP of the finance sectors, GDP of the government sectors, and share of employment of high-tech sectors. This last control is especially important as one may be worried that our results are driven solely by industry dynamics related to few sectors, independent of entrepreneurship.[10] λ_i and δ_t are CZ and time effects, while $\theta_{i,t}$ are CZ-specific linear trends, which allow us to control for unobserved, time-varying factors at the CZ-level such as local economic growth, policy changes, or demographic shifts.

We run the above regression from 1998 to 2014. In all regressions, we cluster standard errors at the CZ level.[11] The interpretation of β is the effect of entrepreneurship on inequality in our regression, which amounts to an assumption of selection on observables: entrepreneurship should be independent of income inequality and incomes, conditional on time-invariant CZ characteristics, year-fixed effects, CZ-specific linear trends, and observable time-varying CZ characteristics.

[8] Guzman and Stern (2020) suggest that most startups experience an exit event within six years after incorporation.

[9] The coefficient then becomes $\beta_{LR} = \sum_{k=0}^{5} \beta_k$.

[10] For instance, one might be concerned that skyrocketing compensation in the high-tech sector may play a fundamental role in skewing the income distribution. We identify technology sectors following the work of Kile and Phillips (2009). These include Drugs, Computer and Office Equipment, Communication Equipment, Electronic Components, Measure and Control Instruments, Medical Instruments, Telephone Communications, Communication Services, Computer Programming, Data Processing, Research and Development, and Testing Services. Results are robust to excluding high-tech employment share from the specification. It has to be noted that our results are robust even if we were to omit this control.

[11] We are unable to cluster standard errors by state because CZs often span two or more states.

The chief threat to identification in this setting arises from unobservable, time-varying characteristics of CZs that drive the relationship between entrepreneurship and inequality. Specifically, it is possible that locational sorting induces this type of endogeneity. If low- and high-income individuals sort differentially into low or high-entrepreneurship areas, a correlation could result between entrepreneurship and income inequality because this locational sorting affects the CZ income distribution. Note that the direction of the endogeneity bias due to this sorting is a priori uncertain. If high-income individuals relocate from low-entrepreneurship CZs to high-entrepreneurship CZs, this would increase income inequality in high-entrepreneurship areas and bias β upward. If instead, low-income individuals relocate from high-entrepreneurship areas to low-entrepreneurship areas, consistently with a gentrification-based story, this would increase inequality in low-entrepreneurship areas and bias β downward. Thus, the sign depends on the relative magnitude of low- and high-income cross-CZ migration flows. Another threat in our empirical setting may be reverse causality, i.e., areas with high inequality may be fertile ground for high-quality entrepreneurship. In the face of this endogeneity, we must find some source of identifying variation that can be leveraged, such as an instrumental-variables setting. We can estimate two-stage least-squares regressions of the form leveraging a shift-share instrument for entrepreneurship (Bartik, 1991). We construct our instrument by interacting the cross-sectional industry share at the CZ level in 1988, i.e., 10 years before the start of our period, with national-level trends in entrepreneurship.

We calculate the first component using the entrepreneurial index split by industry (NAICS 2-digits) in 1988, and we build the second component in our instrument as the growth of the entrepreneurial index for all CZs. For the validity of shift-share instruments, it is essential that either the shift or the share component is exogenous (Goldsmith-Pinkham et al., 2020). In our case, the share component represents the initial industry distribution within each CZ in 1988, while the shift component is the national-level growth rate of entrepreneurship by industry. To ensure the exogeneity of the shift component, we calculate national growth rates by excluding each focal CZ (a leave-one-out procedure). Additionally, we include CZ-specific linear time trends in our analysis, which absorb time-varying factors specific to each CZ (i.e., they account for local economic conditions, demographic changes, policy shifts, or other unobserved factors that evolve differently across CZs over time).

Despite the wide popularity of this type of instrument (e.g., Card, 2001; Autor et al., 2013; Dorn et al., 2020), its validity has been a focal point for numerous scholars, who have proposed various checks and diagnostic tests to ensure their robustness. One prevalent method is the pre-trend test (Goldsmith-Pinkham et al., 2020), which assesses whether the residuals from a second-stage regression

(where the endogenous variable is instrumented using the "current" version of the Bartik instrument) can be predicted by a future version of the instrument. Adao et al. (2019) have advanced this field by introducing two novel inference procedures that generate confidence intervals accommodating a shift-share structure in the residuals. Furthermore, Borusyak et al. (2020) emphasizes the importance of testing the exogeneity of the shock component of the instrument. Our instrument, as detailed in Section A-1 of the Appendix, successfully passes all these recommended checks.

5 Entrepreneurship and Inequality, and Incomes

In this section, we begin by providing some baseline results from fixed-effects models on the correlation between entrepreneurship and the Gini index. We will examine the reason underlying our result, looking at the Lorenz curve. This will be followed by the correlations considering the average income of the median earner (i.e., individuals at the median of the income distribution) as well as other percentile points on the income distribution.

5.1 Entrepreneurship and Inequality

We begin with the first mechanism in our framework, the concentration of returns (Section 2), which predicts that high-quality entrepreneurship widens the income distribution by concentrating gains at the top. Table 2 Column (1) considers the correlation between entrepreneurship and inequality, measured by the Gini index. The resulting estimate suggests a positive relationship between these two variables, even when considering high-growth entrepreneurship, as opposed to more general measures (e.g., Sorensen & Sorenson, 2007; Aghion et al., 2019). The resulting coefficient suggests that an increase in entrepreneurship of 1 point, which roughly corresponds to the establishment of 10 high-quality businesses, that is, incorporated companies operating in a high-tech sector, holding a patent and a trademark, and incorporated in Delaware, is associated with an increase of 0.0011 in the Gini index. For reference, the Gini index increased by 0.015 points from 2000 to 2014 in an average CZ. To place this magnitude in context, the Gini coefficient has a standard deviation of 0.040 across commuting zones and 0.023 within a commuting zone over time, the variation that our fixed-effects estimates exploit. The estimated effect of 0.0011 therefore corresponds to about 2.8% of the cross-zone standard deviation of the Gini, or 4.8% of its within-zone standard deviation, and to roughly 7% of the 0.015-point increase that the average commuting zone experienced over 2000 to 2014. This association also weakens over the sample period: when we estimate the model separately for the two halves of our window, the positive effect on the Gini index is concentrated in the earlier years (1998–2006) and is no longer distinguishable from zero in the later years (2007–2014), a pattern that tracks the secular decline in high-growth entrepreneurship documented over the same period (Decker et al., 2016; Kim

et al., 2026).[12]

At the same time, we might be worried that the positive relationship found in the preceding regression is biased, given the possible endogeneity of our entrepreneurship measures. Thus, we proceed to instrument the measure for entrepreneurship with the instrument described in the previous section. Column (1) of Table 3 shows the correlation between entrepreneurship and our proposed instrument. As expected, the relationship is positive and significant. Overall, the main results of Table 2 are confirmed, with an instrumented coefficient that results only slightly higher than the one resulting from the fixed-effects estimate, but still not significantly different from it. The instrumented coefficient suggests that an increase in entrepreneurship of 1 point is associated with an increase of 0.0013 in the Gini index.[13] We read these instrumented estimates as a way to reduce the endogeneity concerns outlined above and as a confirmation of our fixed-effects results.

We might be interested in understanding the dynamics behind the association between entrepreneurship and higher inequality. To do so, we examine the effect on the Lorenz curve, and we use the Lorenz curve ordinates as dependent variables in our regressions. Lorenz curve ordinates represent the share of income accruing to the bottom x percent of the income distribution, so a marginal effect b on a Lorenz ordinate can be interpreted as a $-b$ effect on the top $1 - x$ percent's share of income. For simplicity, we report coefficients with their sign reversed, so that they reflect an increase of the income share above a particular percentile.

Figure A2 presents the coefficients of a fixed-effects model (left figure) and an instrumented model (right figure) in which entrepreneurship is regressed on every fifth percentile of the income distribution. Both models are aligned in their results and they show that entrepreneurship tends to be associated with an increase in the shares of income toward the top of the distribution. In particular, the instrumented regressions suggest that this effect is positive and significant starting around the 55th percentile, with results that tend to be insignificant before that point.

Another aspect to consider is whether this effect is long-lived or not. The left panel in Figure 4 shows the coefficient of fixed-effects regressions that use different lags (from a cumulative lag of 1 to 10 years) when considering entrepreneurship and the Gini index. The coefficients form an inverted U-shape that culminates around years 5 and 6, when startups are more likely to experience an exit

[12] These sub-period estimates are omitted here for brevity and are available upon request.

[13] Because the specification includes multiple endogenous regressors, inference based on conventional 2SLS standard errors may be sensitive to weak identification. The Anderson Rubin test provides identification robust inference by testing whether the coefficients on the endogenous variables are jointly equal to zero in a way that remains valid even if the instruments are weak. In practice, in order to implement the test with our current setup, we partial out the high-dimensional fixed effects, re-estimate the IV model using the residualized variables, and conduct an Anderson Rubin joint test over all lagged endogenous terms with clustered standard errors. The test strongly rejects the null of no effect, indicating that our conclusions are not driven by weak instrument concerns.

(i.e., M&A or IPO) (Guzman & Stern, 2020), and then decline. This attenuation is encouraging: it suggests that entrepreneurship's upward pressure on inequality is temporary rather than persistent, with inequality effects plateauing and eventually diminishing over time.[14]

5.2 Entrepreneurship and Average Incomes: Who Gains from Entrepreneurship?

We next examine the second mechanism in our framework, labor-market restructuring and diffusion (Section 2), which holds that high-quality entrepreneurship raises incomes across the whole distribution, and not only at the top, through job creation, competition for workers, and spillovers to complementary sectors. Having established a positive relationship between entrepreneurship and inequality, one that seems to taper off over time, we also need to assess whether entrepreneurship affects the income levels of individuals across the income distribution. A positive relationship might partially mitigate the negative effect of entrepreneurship: while such a relationship would be responsible for an unequal distortion of relative incomes in the short run, it might still be able to increase absolute incomes, thus potentially improving the economic conditions of people in a given part of the income distribution. Shedding more light on the relationship between entrepreneurship and average income is another main contribution of this paper as, to our knowledge, this is the first paper studying this relationship.

We first consider the correlation between entrepreneurship and the income of the "median" individual, defined as individuals that belong to the median point of the income distribution. For completeness, we consider both the income in dollars as well as in logs. Columns (2) and (3) in Table 2 show the result of this analysis. Coefficients are positive and significant, highlighting how while entrepreneurship has a positive effect on inequality, it also has a positive effect on income for the median individual. Once again, we are also interested in examining the instrumented results, because of the potential endogeneity problem outlined in the previous sections. Columns (3) to (4) of Table 3 replicate the estimates in Table 2 while instrumenting entrepreneurship. As in the case concerning inequality, coefficients tend to be slightly higher, perhaps consistent with a sorting akin to gentrification-related dynamics. However, once again it is worth noticing that in both cases, the instrumented coefficients are not statistically different from the ones coming from the fixed-effect analyses. The instrumented results suggest that an increase in entrepreneurship of 1 point is associated with an increase of almost 1 percent in the income of the median individual, which translates to an increase of 288 dollars.

[14] One might worry that these results reflect solely founder wealth realization. In analyses not shown here using American Community Survey (ACS) data, which do not include capital gains, we find similarly positive effects and dynamics on inequality and income, consistent with the effect operating through labor market restructuring rather than solely through founder wealth realization.

These results are robust to a series of robustness checks presented in Table A4 in the Appendix. First, we re-ran the analysis without any controls to ensure that our findings are not dependent on the inclusion of control variables. The coefficient remains consistent, indicating that no control is driving the observed relationships. Second, we accounted for the influence of land value and real estate prices to ensure that the results are not driven by landlord gains. We measured this using the land value in USD per acre[15] and the median rent for a two-bedroom dwelling[16]. Third, we included controls for the presence of large establishments[17] that might affect wages and the labor composition in an area, ensuring that our results are not swayed by such factors. Fourth, we controlled for the number of exits, such as IPOs or mergers[18], to determine whether the association between high-quality entrepreneurship and incomes holds even when factoring in exit events explicitly. The results showed that the RECPI index effectively captures high-quality entrepreneurship that does not necessarily result in an IPO, further validating our findings. Additionally, we tested the robustness of our results by excluding CZs located in major entrepreneurial hubs like Massachusetts and California, as well as excluding both simultaneously. Our findings remained robust, demonstrating that the results are not driven by these specific regions. If anything, these trends tend to be even more positive when excluding these areas.

In line with the previous analysis, we also want to check the long-term effect that entrepreneurship might have on the income of the median individual to make sure this is not just a short-term effect. Consistently with the results about inequality, the right panel of Figure 4 shows a steadily increasing trend over time when it comes to the effect of entrepreneurship on income, suggesting that these effects are not short-lived but instead persist and grow over time.

We might be also interested in expanding the analysis beyond the income of the median individual and examining the income of individuals at different points in the income distribution. To do so, we go back to consider our standard time horizon, i.e., 5 years, and we regress the log of the average income of each decile in the income distribution on entrepreneurship. Figure 2 plots the coefficients resulting from this exercise. The average income of the 99th percentile is also reported. The left figure reports fixed effects estimates, while the one on the right reports instrumented estimates. Both sets of results suggest a consistent story: entrepreneurship is associated with higher average incomes across the income distribution, and this pattern seems to be especially pronounced for individuals belonging

[15]Source: United States Department of Agriculture National Agricultural Statistics Service
[16]Source: Housing and Urban Development Health Economic and Residential Stability, 50th Percentile Rent Estimates, 2001-2018
[17]Source: County Business Patterns. Large establishments are defined as establishments employing more than 1,000 employees.
[18]Source: SDC Platinum, Thomson Reuters

to both tails of the income distribution. Instrumented coefficients suggest that an increase of one point in entrepreneurship is associated with an increase of almost 7 percent in the income of individuals in the first decile of the income distribution. This figure goes down to 2 percent for individuals at the second decile, and it increases to 2.5 percent for individuals in the top 1 percent of the income distribution.

However, translating these percentage effects into dollar values reveals that while entrepreneurship's income effects are statistically detectable across the distribution, the economic magnitude of gains is trivial for most individuals and meaningful only for top earners. Over a five-year period, the first decile gains $40, the second decile gains $108, and the median individual gains $288.[19] In stark contrast, the top 1 percent gains ≈$26,300, representing a ratio of approximately 650:1 between top and bottom earners. For most individuals, these gains are economically negligible and do little to improve economic well-being or financial stability. The vast majority of economic benefits from entrepreneurship accrue to those already at the top of the income distribution, with gains concentrated among the top 1 percent. These patterns suggest that entrepreneurship operates primarily as a wealth concentration mechanism rather than a broad-based prosperity engine, despite producing statistically significant effects across the full distribution.

It is also important to note that these effects are not achievable through a more "general" type of entrepreneurship, i.e., subsistence-based entrepreneurship (Schoar, 2010), but are exclusive to high-growth, high-quality entrepreneurship. Figure 3 compares the effects of high-quality entrepreneurship on incomes vis-a-vis the simple number of new businesses (not quality-adjusted). The figure indicates that the positive income effects previously identified are generated specifically by high-growth entrepreneurship, rather than average entrepreneurial endeavors that include subsistence businesses. This divergence is exactly what one would expect if the income and inequality effects operate through the way high-growth firms reshape the local economy. Such firms create jobs, raise the demand for and the returns to labor, so that entrepreneurial activity can substitute for or augment wage income, and they attract outside investment; the marginal business in a simple registration count, which is often a subsistence or self-employment venture (Schoar, 2010), typically does none of these things. The same number of new businesses can therefore carry very different distributional consequences depending on whether those businesses are positioned to grow, which is precisely the dimension that our quality-adjusted measure is built to capture. This result speaks to the importance of distinguishing between types of entrepreneurship, given they

[19]The percentage effects are estimated from specifications using the log of average income as the dependent variable, whereas the dollar figures come from separate specifications estimated in levels.

might have different effects on inequality and incomes.

We also extend our analysis by examining whether the increase in income is driven by specific industries. Figure 5 presents our usual plot, broken down by industry. Generally, all industries show positive effects across most of the income distribution, particularly from the 2nd to the 9th decile. However, results for the tails of the distribution vary in both significance and magnitude. While the right tail generally benefits from the establishment of businesses in all industries, the left tail does not consistently show the same gains. For instance, incomes in the first decile tend to be boosted by the establishment of high-quality businesses that also employ lower-skilled workers. This is evident in industries such as natural resources, manufacturing, transportation, professional services, and leisure and hospitality.

5.2.1 Demographic Characteristics

In light of the results in the previous section, we now delve deeper into understanding which demographics of individuals tend to gain more from entrepreneurship. Due to data limitations, the remainder of the paper will use state-level data rather than CZ-level data. Table A2 in the Appendix reports basic descriptive statistics on incomes for each demographic group we examine. Although this change in geographical scope affects the comparability of coefficient magnitudes across analyses, our baseline estimates at the state level, included in the appendix, yield similar results to those at the CZ level (see Table A8). The CZ-level relationships are also robust to replacing the CZ-specific trends with state-by-year and Census division-by-year fixed effects, retaining the CZ and year fixed effects, so that the effect is identified only from variation across commuting zones within the same state (or division) and year (Table A3 in the Appendix).

We begin our analysis by examining which age groups benefit most from entrepreneurship. Figure 6 illustrates the effect of entrepreneurship on the incomes of individuals across different percentiles by age. Unsurprisingly, young individuals, particularly those aged 25 to 44, gain the most from entrepreneurship across all income percentiles. Individuals aged 45 to 54 also experience some income benefits, though this is less pronounced at the highest income percentiles. Generally, very young individuals (16 to 24 years old) and older individuals (55 to 64 years old) do not significantly benefit from entrepreneurship, as the effects for these age groups tend to be insignificant.

Next, we examine income gains across racial groups (Figure 7). The patterns differ markedly by group and by position in the distribution. Entrepreneurship raises the incomes of low-income American Indian and Alaska Native (AIAN) and Asian individuals, with the gains concentrated in the first decile and fading at higher percentiles. For Black and African American individuals the positive effects appear instead from the middle of the distribution upward, while White individuals gain across

most of the distribution. Hispanic individuals, by contrast, show no significant income gains at any point.

These racial patterns reflect the structural barriers discussed in our theoretical framework. For Black individuals, the finding that positive effects emerge only in upper income deciles may indicate that entrepreneurship benefits primarily those already positioned in higher-wage occupations, while structural barriers, including limited access to capital (Cook et al., 2022), weaker networks (Ruef, 2010), occupational segregation (Fairlie & Robb, 2010), and persistent wage gaps (Chetty et al., 2020), constrain benefits for lower-income Black workers. The lack of significant effects for Hispanic individuals across the static income distribution, despite positive mobility effects documented below, suggests that Hispanic workers may benefit through dynamic channels (moving up over time) rather than through immediate income level increases in a given year. For AIAN and Asian individuals in the first decile, positive effects may reflect employment concentration in specific entrepreneurial sectors or geographic clustering in entrepreneurial hubs.

Also, we examine whether native-born individuals or foreign-born individuals (i.e., immigrants) differentially benefit from entrepreneurship. This distinction is particularly important as immigrants have increasingly played a significant role in the entrepreneurial landscape of host countries, both as employees and as founders of successful firms (Lee & Eesley, 2018; Kerr & Kerr, 2020). Figure 8 presents the estimates considering this distinction. While the estimates for native-born individuals are positive, they are marginally insignificant across all percentiles. In contrast, the situation is notably different for foreign-born individuals, who exhibit positive and significant income gains from the 25th to the 75th percentile of the income distribution. This highlights how immigrants are particularly adept at reaping the benefits of entrepreneurial endeavors, underscoring the crucial role they play in the economic growth and dynamism of the host country.

6 Entrepreneurship and Income Mobility

Entrepreneurship is a fundamentally dynamic process, a concept first underscored by Schumpeter through his theory of creative destruction (Schumpeter, 1942). This corresponds to the third mechanism in our framework, mobility dynamics (Section 2). This inherent dynamism suggests that entrepreneurship may influence not only the level of income and its cross-sectional distribution but also the future distribution of income growth. The importance of studying mobility alongside inequality and incomes has been underscored by several economic scholars (Carroll & Chen, 2016), as the snapshot view of income distribution does not capture the fluidity with which individuals and households move within that distribution over time. Therefore, we now turn our attention to how

entrepreneurship affects income mobility, a crucial aspect of economic dynamics.

Using our standard empirical approach and a five-year window, we focus on income mobility as our main dependent variable. We measure this through the Income Transition Matrix (Kondo et al., 2023), which provides the probability that an individual starting in a given income bin moves to another income bin after five years. The initial and final year income bins are defined across all members within a year and state, rather than being specific to the group of interest. It is important to keep in mind that this measure is a relative measure of mobility as it focuses on the individual's movement within the distribution of income percentiles rather than the absolute change in income levels. For instance, individuals can experience upward absolute mobility through an increase in average incomes, yet still find themselves in a worse position relative to others, resulting in downward relative mobility. The transition matrices are computed using records from individuals who are in the tax sample in the initial and final year. If an individual does not file after five years, they are dropped from the sample. This can occur, for instance, due to unemployment. Leveraging these data, we assess the association between entrepreneurship and the likelihood of individuals moving up or down the income ladder over time in a given state and year.

We begin by examining the entire sample of individuals without considering any specific demographic features. Figure 9 presents a matrix that matches the initial income percentile range in the initial year (on the vertical axis) with the percentile range after five years (on the horizontal axis). Each cell shows the coefficients from a fixed-effect regression where the probability of individuals moving from the Xth percentile range to the Yth percentile range over five years is regressed on our standard measure of entrepreneurship. The cell background color indicates the magnitude and sign of significant coefficients. The figure reveals that an increase in entrepreneurship is associated with a higher probability of individuals initially in the lower than 25th percentile reaching the highest income range (>75th percentile) by 0.1 percent five years later, consistent with entrepreneurship serving as a tool for upward income mobility for less well-off individuals. At the same time, higher entrepreneurship is also associated with a higher likelihood of individuals initially in the 50-75th percentile range falling to the lowest percentile range after five years. For all other combinations of ranges, entrepreneurship does not significantly impact their future mobility probabilities.

Given the importance of understanding mobility patterns, we also examine mobility by focusing on a subset of total income, specifically income *not* strictly derived from wages.[20] Through this

[20] Non-wage income includes all sources of income apart from wages and salaries, such as capital gains, dividends, interest income, business income, and rental income.

analysis, we aim to determine whether mobility strictly occurs through employment or other income sources, which are more relevant for entrepreneurs and investors. Figure A6 in the Appendix shows how mobility trends tend to be more positive but still quite balanced when considering only non-wage income.

The top of the income distribution can become more permeable, with more individuals moving from lower income ranges to the top, while also pulling further away in absolute dollar terms, as we documented in the income analysis. This dynamic reflects Schumpeterian creative destruction: entrepreneurship reshuffles positions within the income distribution without necessarily compressing the overall distribution. Some individuals move upward while others move downward, and the composition of who occupies the top may diversify even as the distance between top and bottom widens. The top 1 percent captures the vast majority of economic gains (≈$26,300 versus $40 for the bottom decile), yet some individuals from lower income ranges reach the top. This represents a reordering of positions rather than a reduction of inequality.

Once again, these mobility trends are exclusive to high-quality entrepreneurship. When considering all types of entrepreneurship (Figure A9), the results are largely insignificant across all combinations of income ranges, indicating how "traditional" or "subsistence" entrepreneurship (Schoar, 2010) does not significantly impact mobility.

6.1 Demographic Characteristics

As usual, we begin our demographic analyses with age. Figure 10 in the Appendix presents the mobility matrix for each age range in our dataset. For the younger segment, individuals aged 16 to 24, entrepreneurship does not offer much income mobility. In fact, individuals in the 50th to 75th percentiles of the income distribution are more at risk of falling to the lowest income range after five years. For individuals aged 25 to 34, entrepreneurship is positively associated with the probability of those in the lower income brackets moving up. This upward mobility is most pronounced when considering the probability of reaching the highest income range (>75th percentile). Additionally, entrepreneurship helps maintain the status quo for some individuals already in the top income range, who are more likely to remain in this range five years later. A similar pattern is observed for individuals aged 35 to 44, where those in the lowest two income ranges experience significant upward mobility five years later. These positive effects tend to taper off as we consider higher age brackets. For individuals aged 45 to 54, there is a positive effect for those in the 25th to 50th percentile range, who are more likely to move up to the 50th to 75th percentile range. Finally, for individuals aged 55 to 64, we do not see much positive effect. In fact, these individuals are more likely to move down the income ladder. This analysis highlights the varied impact of entrepreneurship on income mobility across different age

groups, with younger and middle-aged individuals benefiting more significantly from entrepreneurial activities, while older individuals experience less upward mobility or even downward mobility.

When considering race (Figure 11), the association between entrepreneurship and income mobility varies considerably across groups. Asian individuals exhibit the strongest upward mobility, as nearly all initial income ranges become much more likely to end up higher after five years, with the effect particularly pronounced for the top range. Hispanic and White individuals also benefit, especially those starting in the lowest and middle ranges, who become more likely to reach the highest income range. The picture is less favorable for American Indian and Alaska Native (AIAN) and for Black and African American individuals: while the poorest in these groups have a positive probability of reaching the top, those in the middle and upper income ranges face a meaningful probability of moving down, so that entrepreneurship does not offer them stable upward mobility.

When focusing on mobility derived from income not strictly generated by wages, Figure A7 in the Appendix reveals that the positive mobility for AIAN, Asian, and Hispanic individuals primarily happens through wages. This suggests that these groups benefit from entrepreneurship primarily by being employees of startups or other supporting companies within the ecosystem, rather than through direct ownership. In contrast, the situation for Black individuals reveals a different pattern: when considering only non-wage income, their mobility appears more balanced and aligns more closely with the trends observed for White individuals. This suggests that entrepreneurship-related gains for this group may operate through business income and investments rather than wage employment. This pattern underscores that structural barriers in wage labor markets, including occupational segregation and persistent wage gaps, may limit Black workers' ability to capture entrepreneurship's benefits through employment channels, even in regions with high entrepreneurial activity. High-growth startups often hire through personal networks (Ruef, 2010), potentially excluding Black workers from the equity premiums and wage gains that accrue to early technical hires. Occupational segregation may further compound this dynamic by channeling Black workers into support rather than technical roles, where exposure to entrepreneurial upside through wages is more limited.

We then turn our attention to native-born versus foreign-born individuals, as shown in Figure 12. For native-born individuals, the results are very similar to the results found in Figure 9.[21] Foreign-born individuals exhibit in general an exceptional upward mobility, with patterns similar to those observed for Hispanic and Asian individuals. Across all initial income ranges, foreign-born individuals now have higher chances of ending up in a higher income range, indicating that entrepreneurship significantly

[21] As natives constitute most of the population, it makes sense that patterns observed in the broader population largely reflect the experiences of native-born individuals.

benefits their economic mobility. When examining whether their mobility happens through wages or non-wage income, Figure A8 in the Appendix shows that when exclusively considering non-wage income, foreign-born economic mobility does not appear to be as overwhelmingly positive as in the figure considering all income. This indicates that foreign-born individuals also benefit significantly from wage incomes. Typically, immigrants face substantial restrictions, as the majority enter the U.S. on non-immigrant visas.[22]. These visas typically allow them to work only as employees, making it challenging to start their own companies until they obtain permanent residency (Kerr & Kerr, 2020). This constraint limits their opportunities for entrepreneurship and underscores the importance of wage income for their economic mobility.

7 Conclusion

Using new aggregate measures of incomes produced from the universe of tax returns linked with survey data, we have surfaced new evidence on an old question, *Who benefits from creative destruction?*, which we capture with new measures of high-quality entrepreneurship. We shed light on this question by building a new dataset at the microgeographical level, leveraging the universe of IRS Form 1040 tax filings, that includes measures of inequality and average incomes at various points along the income distribution. Unlike other data used in the literature, these data do not suffer from top-income censoring and so offer a faithful representation of incomes even at the extreme right tail.

Our central finding is that high-quality entrepreneurship is associated with higher income inequality, and that its income gains, while statistically detectable across the entire distribution, are economically meaningful almost only at the top. Expressed in dollars, the gap is stark: over five years the top 1 percent gains on the order of $26,300 against roughly $40 for the bottom decile, a ratio of about 650 to 1. For most individuals the gains are negligible in absolute terms.

Entrepreneurship also reshuffles positions within the income distribution. It can lift individuals out of the lowest income bracket, but it carries risk for those in the middle, who face a higher chance of falling. These mobility gains are uneven across groups: they are pronounced for White, Hispanic, Asian, and foreign-born individuals, but weaker or even negative for American Indian and Black individuals. For several groups the gains run mainly through wage income rather than business ownership, and for Black individuals in particular it is wage dynamics that constrain mobility.

Throughout, these patterns are specific to high-quality, high-growth entrepreneurship and do not appear when entrepreneurship is measured by a simple count of new business registrations. This distinction matters for how our results should be read: it is not the sheer volume of new businesses

[22]For more detailed statistics, see: https://travel.state.gov/content/dam/visas/Statistics/AnnualReports/FY2023AnnualReport/FY2023_AR_TableI.pdf

that reshapes the local income distribution, but the presence of firms with genuine growth potential, which create jobs, bid up the returns to labor, and draw in investment. A quantity-based measure that gave equal weight to subsistence and self-employment ventures would have masked these effects, and we see the quality-adjusted measure as essential to recovering them.

Addressing the research questions in this paper is challenging from both empirical and data perspectives, and our study has limitations. Although we use high-quality administrative data to construct detailed aggregates of local income distributions, we cannot fully disentangle the mechanisms through which entrepreneurship affects different groups. Doing so would require linking company-level data to individual IRS filings, which is beyond the scope of the paper. Therefore, we cannot determine whether income gains disproportionately benefit entrepreneurs or investors and how these roles intersect with demographic characteristics. IRS data only capture tax filers; our "bottom decile" represents the bottom of the filing population rather than the absolute poorest individuals. Mobility analyses use state-level data, which may mask important within-state heterogeneity in entrepreneurial activity; because we vary the geographic unit of analysis, we interpret mobility results as associations rather than causal. Additionally, our study focuses exclusively on the United States, characterized by a dynamic entrepreneurial environment and favorable equity market. Investigating whether our findings hold in other countries and contexts would provide insights into the generalizability of our conclusions. Finally, while we document marked differences across demographic groups, and particularly across racial groups, our data let us describe these patterns but not explain why they arise; uncovering their underlying causes is an important question that we leave to future research.

A central rationale for promoting entrepreneurship is the expectation that its benefits are broadly distributed and therefore justify substantial public and private investment. Our findings invite a more qualified view of that premise: entrepreneurship's gains are substantial in the aggregate, but they are concentrated and unevenly shared, so the familiar image of entrepreneurship as an engine of broadly shared prosperity stands in tension with the evidence. While entrepreneurship clearly creates value, the aggregate picture can hide the question at the center of this paper, who actually captures the gains. Because its gains flow through high-growth firms whose opportunities are unevenly within reach, the distributional consequences of entrepreneurship are shaped as much by who can access it as by how much of it occurs. Entrepreneurship can thus be a pathway to the "American Dream" for some, while remaining an elusive goal for many.

References

Acemoglu, D. (2002). Directed technical change. *Review of Economic Studies*, *69*(4), 781–809.

Adao, R., Kolesár, M., & Morales, E. (2019). Shift-share designs: Theory and inference. *The Quarterly Journal of Economics*, *134*(4), 1949–2010.

Aghion, P., Akcigit, U., Bergeaud, A., Blundell, R., & Hémous, D. (2019). Innovation and top income inequality. *The Review of Economic Studies*, *86*(1), 1–45.

Alesina, A., & Rodrik, D. (1994). Distributive politics and economic growth. *The quarterly journal of economics*, *109*(2), 465–490.

Andrews, M. J., Fazio, C., Guzman, J., Liu, Y., & Stern, S. (2022). The role of firm heterogeneity and dynamics in driving aggregate productivity. *Research Policy*, *51*(8), 104515.

Andrews, R., Fazio, C., Guzman, J., Liu, Y., & Stern, S. (2019). The Startup Cartography Project.

Åstebro, T., Chen, J., & Thompson, P. (2011). Stars and misfits: Self-employment and labor market frictions. *Management Science*, *57*(11), 1999–2017.

Autor, D. H., Dorn, D., & Hanson, G. H. (2013). The china syndrome: Local labor market effects of import competition in the united states. *American Economic Review*, *103*(6), 2121–68.

Autor, D. H., & Handel, M. J. (2013). Putting tasks to the test: Human capital, job tasks, and wages. *Journal of Labor Economics*, *31*(S1), S59–S96.

Babina, T., Ma, W., Moser, C., Ouimet, P., & Zarutskie, R. (2019). *Pay, employment, and dynamics of young firms* (tech. rep. No. 19-25). Kenan Institute Research Paper. https://ssrn.com/abstract=3425596

Bartik, T. J. (1991). Who benefits from state and local economic development policies?

Bates, T. (1997). Race, self-employment, and upward mobility: An illusive american dream. *The Woodrow Wilson Center Press*.

Bell, A., Chetty, R., Jaravel, X., Petkova, N., & Van Reenen, J. (2019). Who becomes an inventor in america? the importance of exposure to innovation. *Quarterly Journal of Economics*, *134*(2), 647–713.

Berlingieri, G., Blanchenay, P., & Criscuolo, C. (2024). The great divergence (s). *Research Policy*, *53*(3), 104955.

Binder, D., & Kovacevic, M. (1995). Estimating some measures of income inequality from survey data: An application of the estimating equations approach. *Survey Methodology*, *21*.

Borusyak, K., Hull, P., & Jaravel, X. (2020). Quasi-experimental shift-share research designs. *Review of Economic Studies (forthcoming)*.

Braguinsky, S., Klepper, S., & Ohyama, A. (2012). High-tech entrepreneurship. *Journal of Law and Economics*, *55*(4), 869–900.

Brandtner, C., Sorenson, O., & Feldman, M. (2025). Prosperous places: Processes, policies, and practices. *Industrial and Corporate Change*, *35*(1), 186–205.

Cajner, T., Radler, T., Ratner, D., & Vidangos, I. (2017). Racial gaps in labor market outcomes in the last four decades and over the business cycle. *Federal Reserve Board*.

Card, D. (2001). Immigrant inflows, native outflows, and the local labor market impacts of higher immigration. *Journal of Labor Economics*, *19*(1), 22–64.

Carnahan, S., Agarwal, R., & Campbell, B. A. (2012). Heterogeneity in turnover: The effect of relative compensation dispersion of firms on the mobility and entrepreneurship of extreme performers. *Strategic Management Journal*, *33*(12), 1411–1430.

Carroll, D. R., & Chen, A. (2016). Income inequality matters, but mobility is just as important. *Economic Commentary*, (2016-06).

Chetty, R., Hendren, N., Jones, M. R., & Porter, S. R. (2020). Race and economic opportunity in the united states: An intergenerational perspective. *Quarterly Journal of Economics*, *135*(2), 711–783.

Chetty, R., Hendren, N., Kline, P., & Saez, E. (2014). Where is the land of opportunity? the geography of intergenerational mobility in the united states. *The Quarterly Journal of Economics*, *129*(4), 1553–1623.

Combes, P.-P., Duranton, G., & Gobillon, L. (2010). The identification of agglomeration economies. *Journal of Economic Geography*, *11*(2), 253–266.

Cook, L. D., Marx, M., & Yimfor, E. (2022). *Funding black high-growth startups* (tech. rep.). National Bureau of Economic Research.

Corak, M. (2013). Income inequality, equality of opportunity, and intergenerational mobility. *Journal of Economic Perspectives*, *27*(3), 79–102.

Decker, R., Haltiwanger, J., Jarmin, R., & Miranda, J. (2014). The role of entrepreneurship in us job creation and economic dynamism. *Journal of Economic Perspectives*, *28*(3), 3–24.

Decker, R. A., Haltiwanger, J., Jarmin, R. S., & Miranda, J. (2016). Where has all the skewness gone? The decline in high-growth (young) firms in the U.S. *European Economic Review*, *86*, 4–23.

Delgado, M., Porter, M. E., & Stern, S. (2010). Clusters and entrepreneurship. *Journal of economic geography*, *10*(4), 495–518.

Dorn, D., Hanson, G., Majlesi, K., et al. (2020). Importing political polarization? the electoral consequences of rising trade exposure. *American Economic Review*, *110*(10), 3139–83.

Dube, A. (2019). Minimum wages and the distribution of family incomes. *American Economic Journal: Applied Economics*, *11*(4), 268–304.

Eeckhout, J., Pinheiro, R., & Schmidheiny, K. (2014). Spatial sorting. *Journal of Political Economy*, *122*(3), 554–620.

Fairlie, R. (2018). Racial inequality in business ownership and income. *Oxford Review of Economic Policy*, *34*(4), 597–614.

Fairlie, R. W., Kroff, Z., Miranda, J., & Zolas, N. (2023). *The promise and peril of entrepreneurship: Job creation and survival among us startups*. MIT Press.

Fairlie, R. W., & Robb, A. M. (2010). *Race and entrepreneurial success: Black-, asian-, and white-owned businesses in the united states*. MIT Press.

Frank, R. H., & Cook, P. (1996). *The winner-take-all society: Why the few at the top get so much more than the rest of us*.

Galor, O., & Moav, O. (2004). From physical to human capital accumulation: Inequality and the process of development. *The Review of Economic Studies*, *71*(4), 1001–1026.

Glaeser, E. L., Kerr, S. P., & Kerr, W. R. (2015). Entrepreneurship and urban growth: An empirical assessment with historical mines. *Review of Economics and Statistics*, *97*(2), 498–520.

Goldsmith-Pinkham, P., Sorkin, I., & Swift, H. (2020). Bartik instruments: What, when, why, and how. *American Economic Review*, *110*(8), 2586–2624.

Guzman, J., & Kacperczyk, A. O. (2019). Gender gap in entrepreneurship. *Research Policy*, *48*(7), 1666–1680.

Guzman, J., & Stern, S. (2020). The state of american entrepreneurship: New estimates of the quantity and quality of entrepreneurship for 32 us states, 1988–2014. *American Economic Journal: Economic Policy*, *12*(4), 212–243.

Haltiwanger, J., Jarmin, R. S., & Miranda, J. (2013). Who creates jobs? small versus large versus young. *Review of Economics and Statistics*, *95*(2), 347–361.

Kerr, S. P., & Kerr, W. (2020). Immigrant entrepreneurship in america: Evidence from the survey of business owners 2007 & 2012. *Research Policy*, *49*(3), 103918.

Kerr, S. P., & Kerr, W. R. (2016). Immigrant entrepreneurship. *NBER working paper*.

Kile, C. O., & Phillips, M. E. (2009). Using industry classification codes to sample high-technology firms: Analysis and recommendations. *Journal of Accounting, Auditing & Finance*, *24*(1), 35–58.

Kim, J. D., Choi, J., Goldschlag, N., & Haltiwanger, J. C. (2026). Firm growth and stagnation in the United States: Key trends and new data opportunities. *Strategic Management Journal*, *47*(4), 1039–1065.

Kim, J. D. (2018). Is there a startup wage premium? evidence from mit graduates. *Research Policy*, *47*(3), 637–649.

Kondo, I., Rinz, K., Hawkins, B., Gubbay, N., Voorheis, J., & Wozniak, A. (2023). *Granular income inequality and mobility using idda* (Working Paper CES-23-55). U.S Census Bureau, v.1.0.

Kwon, D., & Sorenson, O. (2021). The silicon valley syndrome. *Entrepreneurship Theory and Practice*, *47*(2), 344–368.

Lee, Y. S. (2017). Entrepreneurship, small businesses and economic growth in cities. *Journal of Economic Geography*, *17*(2), 311–343.

Lee, Y. S., & Eesley, C. (2018). The persistence of entrepreneurship and innovative immigrants. *Research Policy*, *47*(6), 1032–1044.

Lieberman, M. B., & Montgomery, D. B. (1988). First-mover advantages. *Strategic management journal*, *9*(S1), 41–58.

Lu, J. G., Nisbett, R. E., & Morris, M. W. (2020). Why east asians but not south asians are underrepresented in leadership positions in the united states. *Proceedings of the National Academy of Sciences*, *117*(9), 4590–4600.

Marshall, A. (1920). Principles of economics. *London: Mac-Millan*, 1–627.

Moretti, E. (2010). Local multipliers. *American Economic Review*, *100*(2), 373–377.

Moretti, E. (2012). *The new geography of jobs*. Houghton Mifflin Harcourt.

Neumark, D., Wall, B., & Zhang, J. (2011). Do small businesses create more jobs? new evidence for the united states from the national establishment time series. *The Review of Economics and Statistics*, *93*(1), 16–29.

Ouimet, P., & Zarutskie, R. (2014). Who works for startups? the relation between firm age, employee age, and growth. *Journal of financial Economics*, *112*(3), 386–407.

Piketty, T., & Saez, E. (2003). Income inequality in the united states, 1913-1998. *Quarterly Journal of Economics*, *118*(1).

Romer, P. M. (1990). Endogenous technological change. *Journal of Political Economy*, *98*(5), S71–S102.

Ruef, M. (2010). *The entrepreneurial group: Social identities, relations, and collective action*. Princeton University Press.

Saxenian, A. (1994). Regional networks: Industrial adaptation in silicon valley and route 128.

Schmieder, J. (2023). Establishment age and wages. *Journal of Econometrics*, *233*(2), 424–442.

Schoar, A. (2010). The Divide between Subsistence and Transformational Entrepreneurship. *Innovation Policy and the Economy*, *10*(1), 57–81.

Schumpeter, J. (1942). Creative destruction. *Capitalism, socialism and democracy*, *825*.

Sorensen, J. B., & Sharkey, A. J. (2014). Inequality in social capital: Evidence from entrepreneurship. *American Sociological Review*, *79*(3), 432–454.

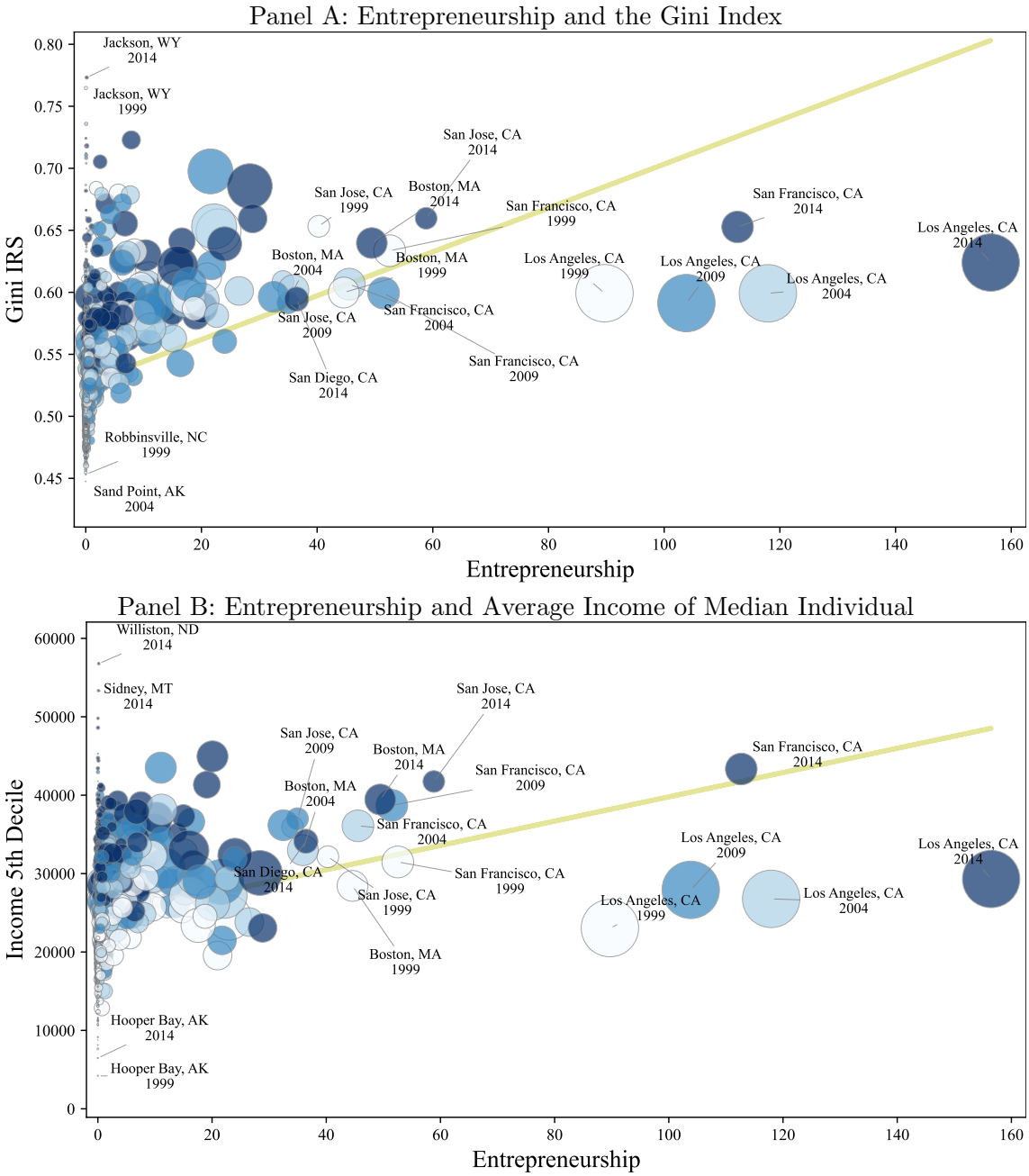
Sorensen, J. B., & Sorenson, O. (2007). Corporate Demography and Income Inequality. *American Sociological Review*, *72*(5), 766–783.

Tolbert, C. M., & Sizer, M. (1996). *Us commuting zones and labor market areas: A 1990 update* (Staff Report AGES-9614). United States Department of Agriculture, Economic Research Service.

Van Reenen, J. (1996). The creation and capture of rents: Wages and innovation in a panel of uk companies. *Quarterly Journal of Economics*, *111*(1), 195–226.

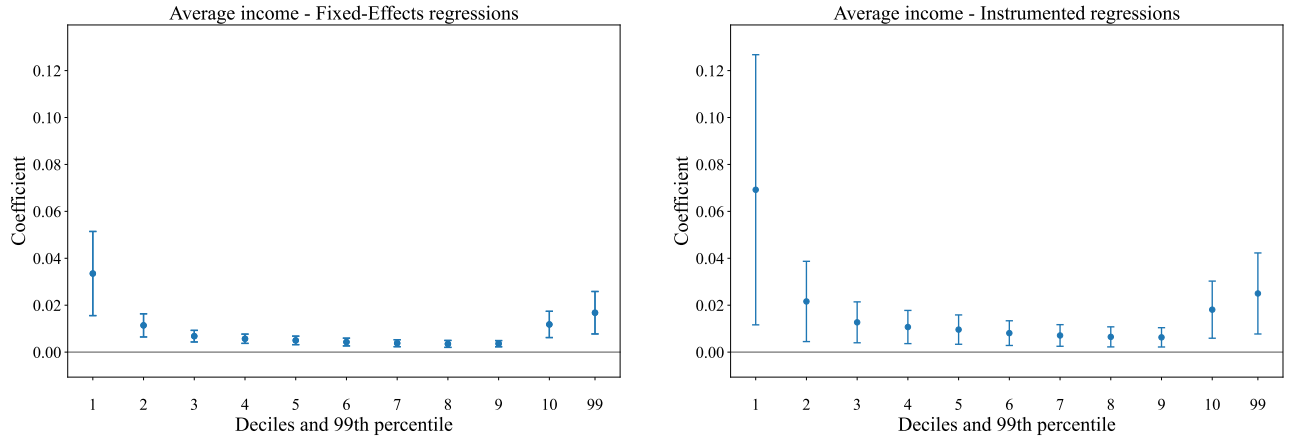
Figures and Tables

Figure 1: Scatterplot of Entrepreneurship, the Gini Index, and Average Income for the Median Individual for Selected Years



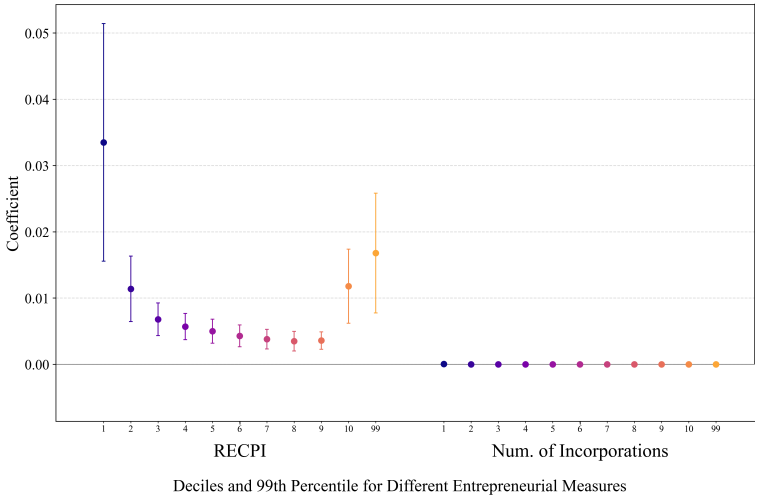
Sources: IRS Form 1040s, 1998–2014; Andrews et al. (2019).
Notes: Darker dots are associated with more recent years. For clarity, we consider only the years 1999, 2004, 2009, and 2014. Dot size is proportional to the population of the CZ in 2014. DRB approval numbers CBDRB-FY18-164, CBDRB-FY19-342, and CBDRB-FY19-509.

Figure 2: Coefficients of FE and IV Regression of Average Income by Deciles and 99th Percentile on Entrepreneurship



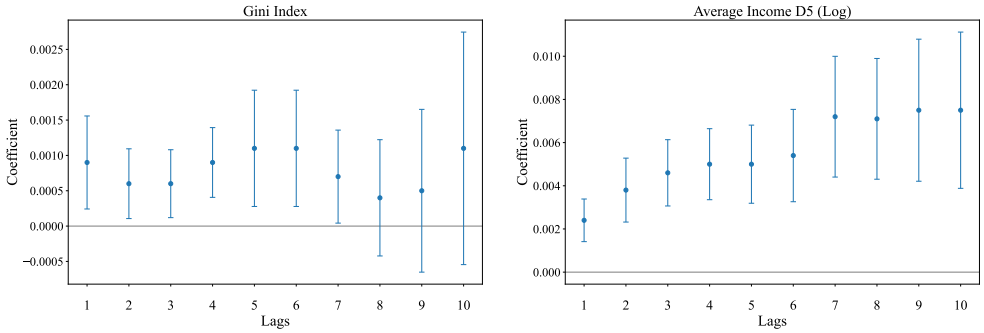
Sources: IRS Form 1040s, 1998–2014; Andrews et al. (2019).
Notes: The graphs present coefficients of several fixed-effects (left graph) and instrumented (right graph) regressions where different levels of average income (which include all the deciles and the top 1 percent of the income distribution) are regressed on entrepreneurship. All regressions include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GDP of the finance sectors, GDP of the government sectors. Standard errors are clustered at the CZ level. The length of the vertical segments indicates 10-percent confidence-level intervals. IV regressions for which the F-statistic of the first stage is less than 10 are presented as dashed standard error lines. DRB approval numbers CBDRB-FY18-164, CBDRB-FY19-342, and CBDRB-FY19-509.

Figure 3: Coefficients of FE Regression of Average Income by Deciles and 99th Percentile on Different Entrepreneurial Measures



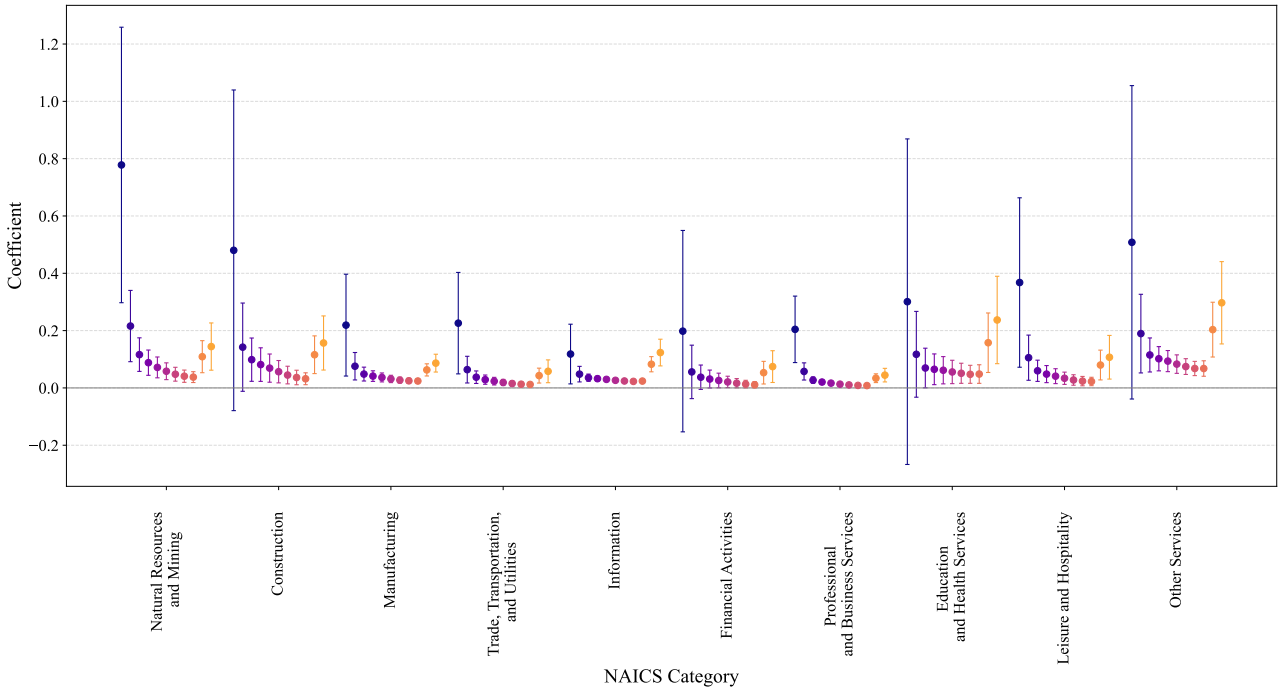
Notes: The graphs present coefficients of several fixed-effects regressions where the average income of different income percentiles are regressed on several measures capturing entrepreneurship. In particular, we consider the RECPI index, the preferred entrepreneurship measure used in this paper, and the simple number of incorporations. All regressions include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GDP of the finance sectors, and GDP of the government sectors. Standard errors are clustered at the CZ level. The length of the vertical segments indicates 10-percent confidence-level intervals. DRB approval numbers CBDRB-FY18-164, CBDRB-FY19-342, and CBDRB-FY19-509.

Figure 4: Coefficients of FE Regression of the Gini Index and the Average Income of Individuals at the Median of the Income Distribution (short- and long-term effects)



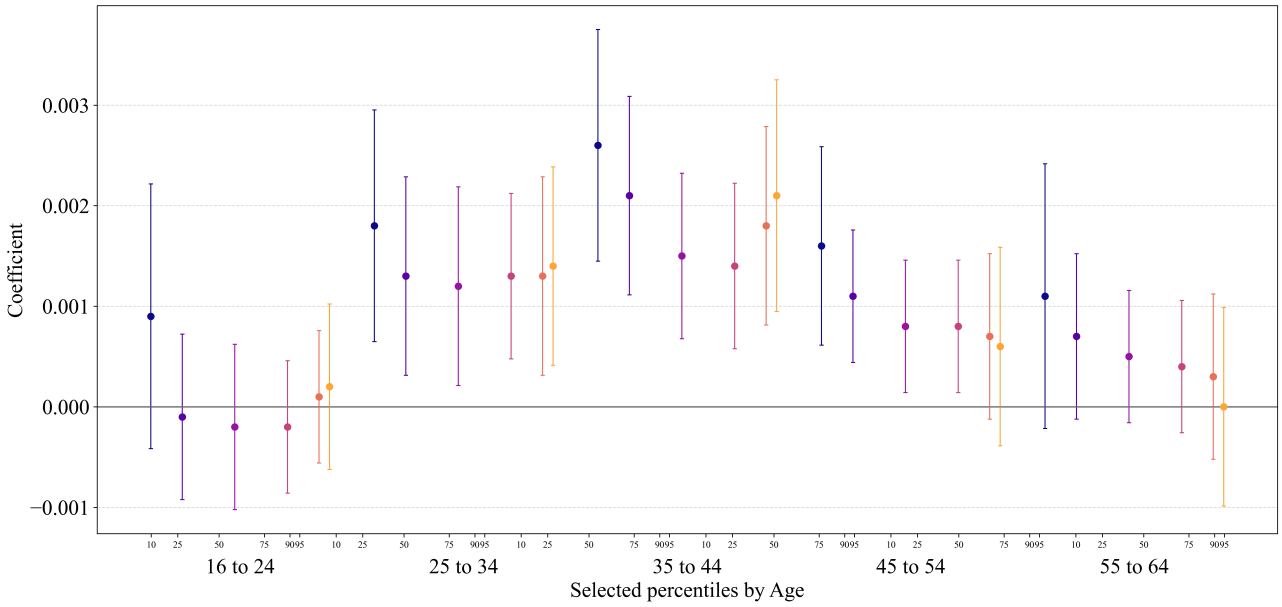
Notes: The graphs present coefficients of several fixed-effects regressions where the Gini Index (left figure) and the average income of individuals at the median of the income distribution (right figure) are regressed on entrepreneurship using different lags (1 to 10 years). All regressions include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GDP of the finance sectors, and GDP of the government sectors. Standard errors are clustered at the CZ level. The length of the vertical segments indicates 10-percent confidence-level intervals. DRB approval numbers CBDRB-FY18-164, CBDRB-FY19-342, and CBDRB-FY19-509.

Figure 5: Coefficients of FE Regression of Average Income by Deciles and 99th Percentile by Industry



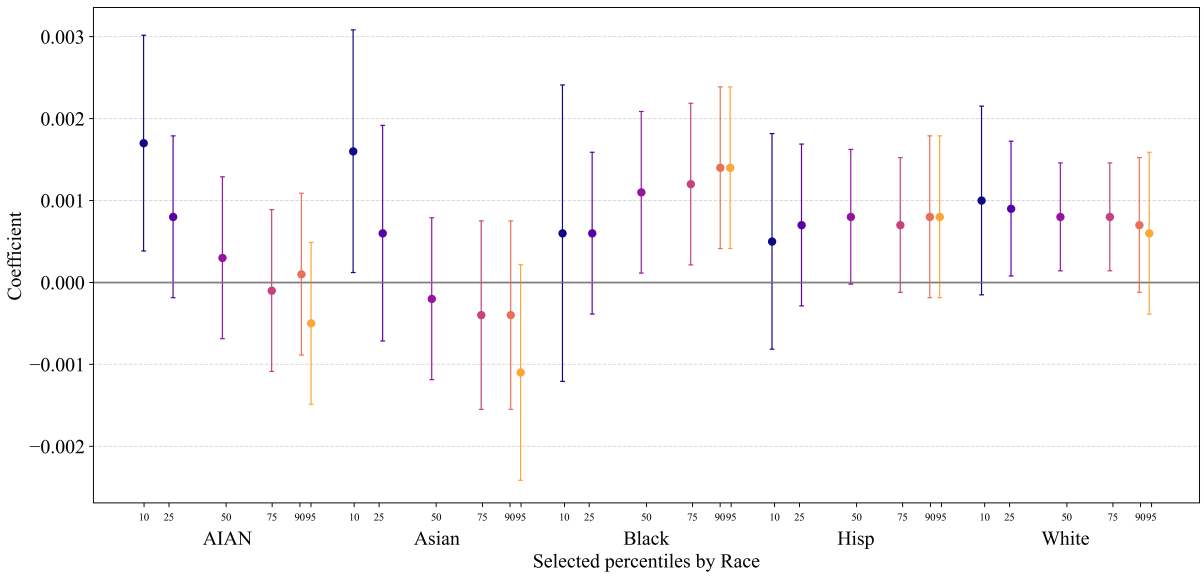
Notes: The graphs present coefficients of several fixed-effects regressions where the average income of different income percentiles are regressed on entrepreneurship by industry. All regressions include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GDP of the finance sectors, and GDP of the government sectors. Standard errors are clustered at the CZ level. The length of the vertical segments indicates 10-percent confidence-level intervals. DRB approval numbers CBDRB-FY18-164, CBDRB-FY19-342, and CBDRB-FY19-509.

Figure 6: Coefficients of FE Regression of Average Income by Deciles and 99th Percentile by Age



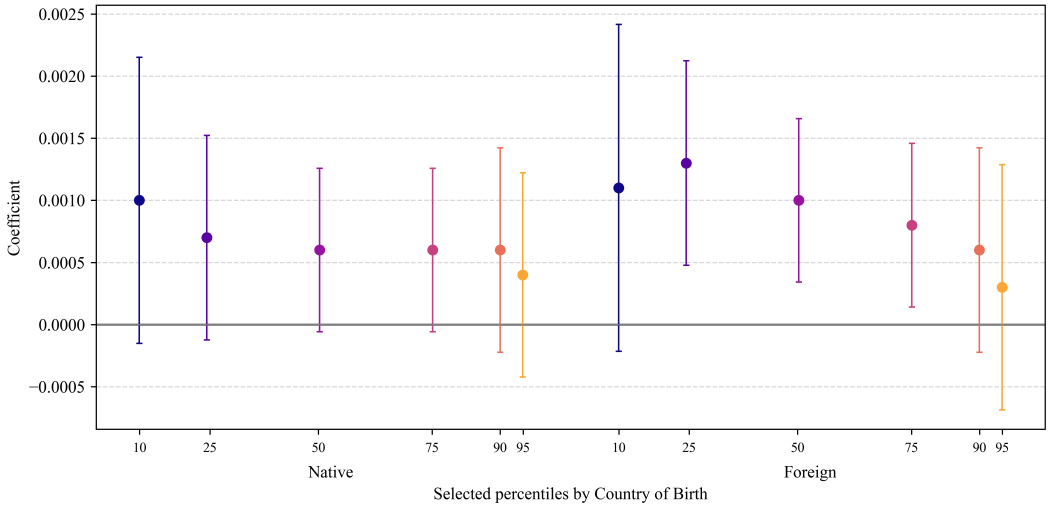
Notes: The graphs present coefficients of several fixed-effects regressions where the average incomes of different income percentiles are regressed on entrepreneurship by age. All regressions include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GDP of the finance sectors, and GDP of the government sectors. Standard errors are clustered at the CZ level. The length of the vertical segments indicates 10-percent confidence-level intervals.

Figure 7: Coefficients of FE Regression of Average Income by Deciles and 99th Percentile by Race



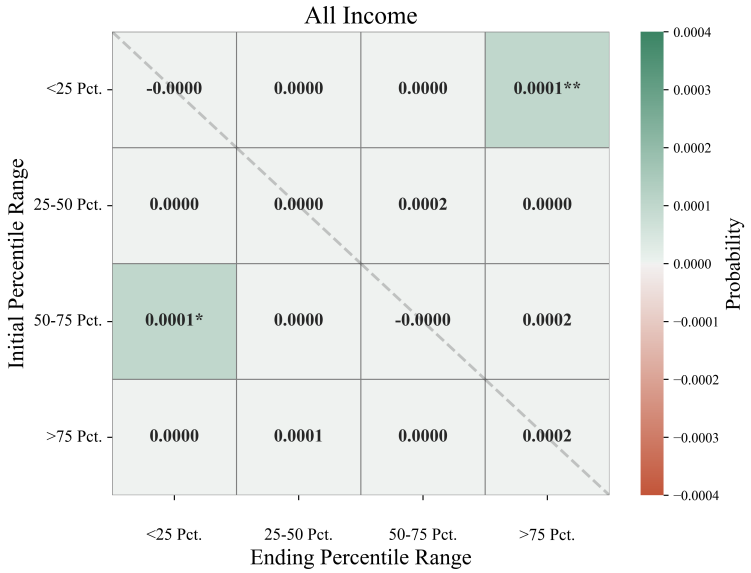
Notes: The graphs present coefficients of several fixed-effects regressions where the average incomes of different income percentiles are regressed on entrepreneurship by race. All regressions include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GDP of the finance sectors, and GDP of the government sectors. Standard errors are clustered at the CZ level. The length of the vertical segments indicates 10-percent confidence-level intervals.

Figure 8: Coefficients of FE Regression of Average Income by Deciles and 99th Percentile by Country of Birth



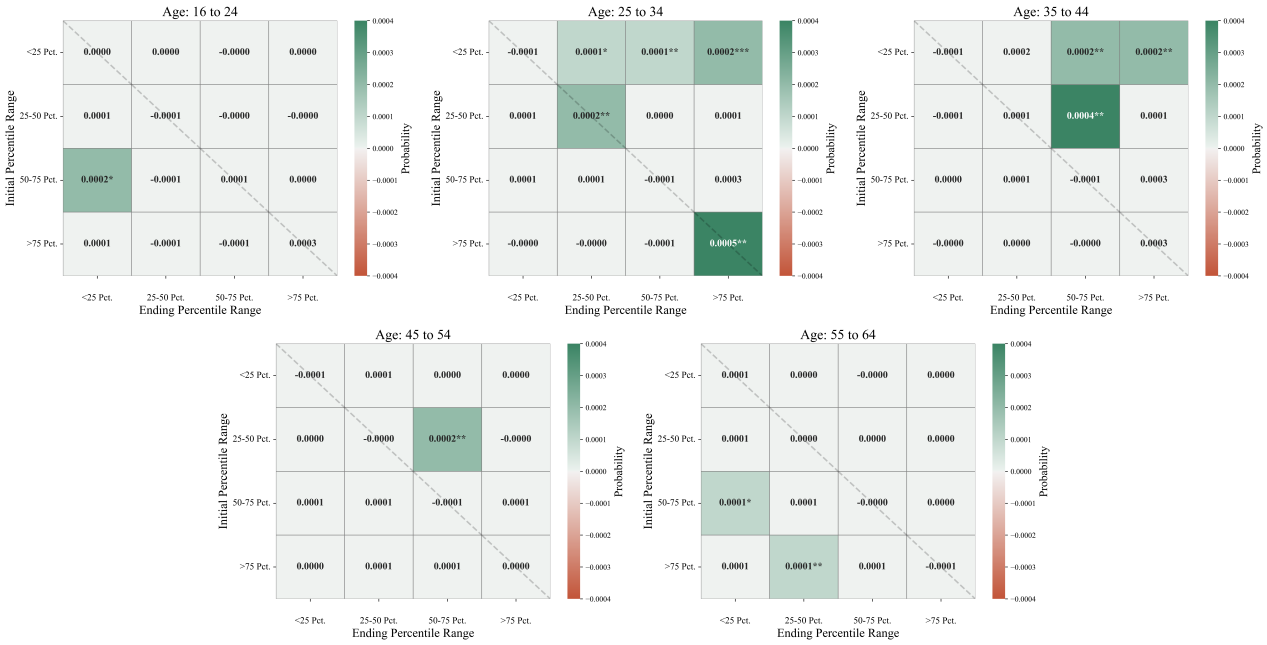
Notes: The graphs present coefficients of several fixed-effects regressions where the average incomes of different income percentiles are regressed on entrepreneurship by country of birth. All regressions include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GDP of the finance sectors, and GDP of the government sectors. Standard errors are clustered at the CZ level. The length of the vertical segments indicates 10-percent confidence-level intervals.

Figure 9: Coefficients of FE Regressions of the Probability of belonging to a given Percentile Range by Initial Percentile Range after 5 years



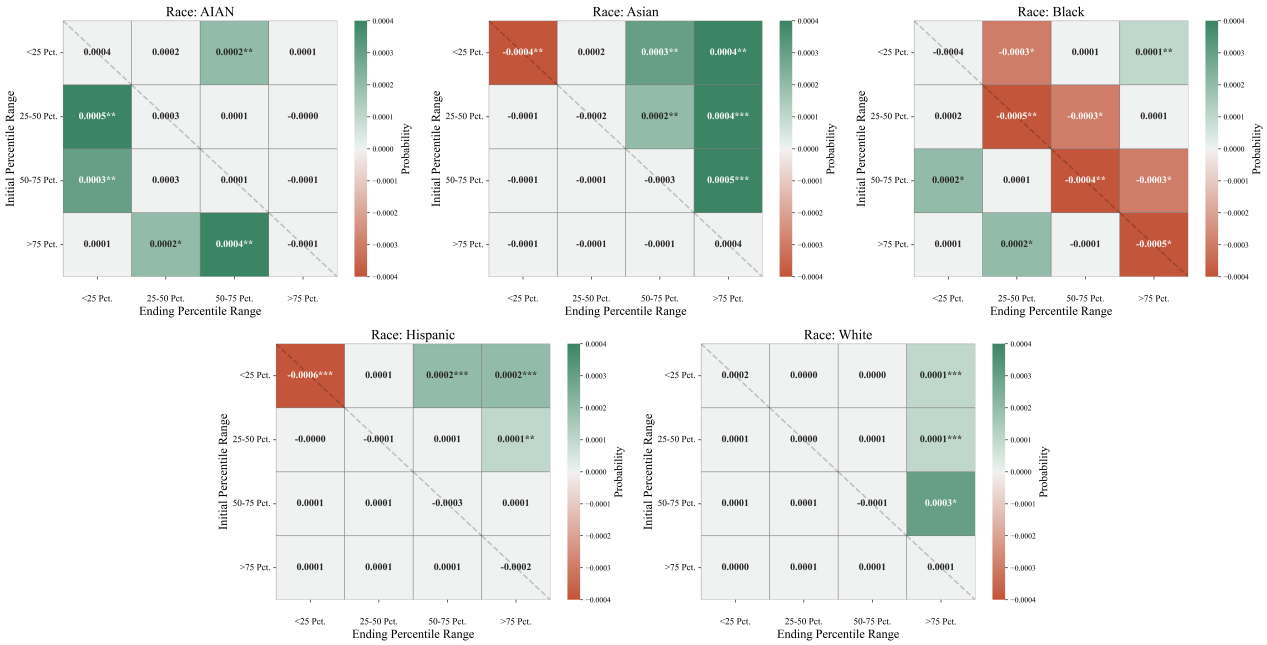
Sources: Kondo et al. (2023); Andrews et al. (2019). *Notes:* The graphs present coefficients of several fixed-effects regressions where measures of relative mobility (i.e., the probability of individuals belonging to the Xth percentile to belong to the Yth percentile after 5 years) are regressed on high-quality entrepreneurship. All regressions include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GDP of the finance sectors, and GDP of the government sectors. Standard errors are clustered at the state level.

Figure 10: Coefficients of FE Regressions of the Probability of belonging to a given Percentile Range by Initial Percentile Range after 5 years by Age



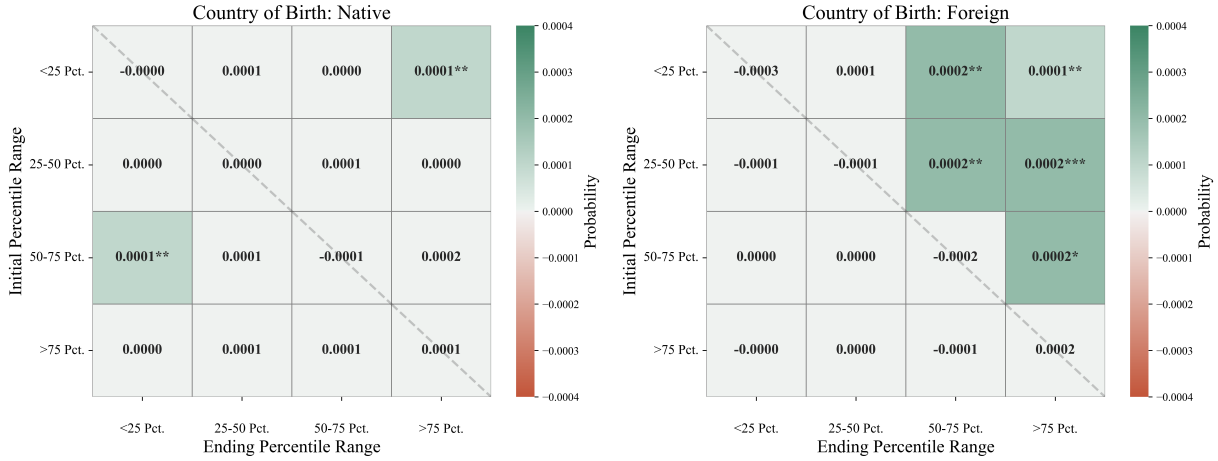
Sources: Kondo et al. (2023); Andrews et al. (2019). *Notes:* The graphs present coefficients of several fixed-effects regressions where measures of relative mobility (i.e., the probability of individuals belonging to the Xth percentile to belong to the Yth percentile after 5 years) are regressed on high-quality entrepreneurship by age. All regressions include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GDP of the finance sectors, and GDP of the government sectors. Standard errors are clustered at the state level.

Figure 11: Coefficients of FE Regressions of the Probability of belonging to a given Percentile Range by Initial Percentile Range after 5 years by Race



Sources: Kondo et al. (2023); Andrews et al. (2019). *Notes:* The graphs present coefficients of several fixed-effects regressions where measures of relative mobility (i.e., the probability of individuals belonging to the Xth percentile to belong to the Yth percentile after 5 years) are regressed on high-quality entrepreneurship by race. All regressions include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GDP of the finance sectors, and GDP of the government sectors. Standard errors are clustered at the state level.

Figure 12: Coefficients of FE Regressions of the Probability of belonging to a given Percentile Range by Initial Percentile Range after 5 years by Country of Birth



Sources: Kondo et al. (2023); Andrews et al. (2019). *Notes:* The graphs present coefficients of several fixed-effects regressions where measures of relative mobility (i.e., the probability of individuals belonging to the Xth percentile to belong to the Yth percentile after 5 years) are regressed on high-quality entrepreneurship by country of birth. All regressions include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GDP of the finance sectors, GDP of the government sectors. Standard errors are clustered at the state level.

Table 1: Descriptive statistics of the main dependent and independent variables

	N	Mean	Median	St.Dev.	Min	Max
Gini Index	12036	0.53	0.52	0.04	0.43	0.94
Avr. Income P10 ($000's)	12036	2.6	2.7	0.8	0.0	6.3
Avr. Income P50 ($000's)	12036	24.0	23.6	4.9	3.9	56.8
Avr. Income P90 ($000's)	12036	173.0	157.1	82.5	56.1	4740.4
Entrepreneurship	12019	0.99	0.05	5.96	0.00	156.45
Population (count)	12048	419913	118647	1099840	946	18559545
Unemployment (rate)	11747	0.05	0.05	0.03	0.00	0.27
GDP ($M)	11747	14272	3492	39807	0	856162
Asian (count)	12048	21189	895	122241	0	2689100
Black (count)	12048	56327	3717	187661	0	2780954
Hispanic (count)	12048	56900	4108	319195	0	7891695
Under 25 (count)	12048	116398	32863	308192	194	5285244
Over 65 (count)	12048	54136	17863	128928	46	2291204
Female (count)	12048	213610	60206	561019	421	9388494

Sources: IRS 1040s; Andrews et al. (2019). *Notes:* Statistics at the CZ-year level for the period 1998–2014.

Table 2: Fixed-Effects Regressions of Entrepreneurship on the Gini index and Average Income of Individuals at the Median of the Income Distribution

	(1)	(2)	(3)
Model	FE	FE	FE
Dep. Var.	Gini index	Avg. Income 5th Decile (USD)	Avg. Income 5th Decile (Log)
Entrepreneurship	0.0011**	152.3***	0.0050***
	(0.0005)	(34.6)	(0.0011)
N	11,695	11,695	11,695

Sources: IRS 1040s, 1998–2014; Andrews et al. (2019). *Notes:* Estimates of fixed-effects regressions. The dependent variable in column 1 is the Gini index, while in columns 2 and 3 the dependent variables are the average income of individuals located at the median point of the income distribution in USD and logged values, respectively. All columns include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GPD of the finance sectors, and GDP of the government sectors. All variables are at the Commuting Zone level and include Commuting Zone fixed-effects, year fixed-effects, and Commuting Zone-year trends. Errors are clustered at the Commuting Zone level: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Instrumented Regressions of Entrepreneurship on the Gini index and Average Income of Individuals at the Median of the Income Distribution

	(1)	(2)	(3)	(4)
Model	First Stage	IV	IV	IV
Dep. Var.	Entrepreneurship	Gini index	Avg. Income 5th Decile (USD)	Avg. Income 5th Decile (Log)
Shift-Share IV	3.745***			
	(0.405)			
Entrepreneurship		0.0013**	288.0**	0.0096**
		(0.0006)	(112.5)	(0.0038)
Observations	11,712	11,695	11,695	11,695
F-stat		1560.63	1560.63	1560.63
KP F-stat		13.39	13.39	13.39

Sources: IRS 1040s, 1998–2014; Andrews et al. (2019). *Notes:* Estimates of instrumented fixed-effects regressions where entrepreneurship (RECPI index) is instrumented with a shift-share instrument. Column 1 shows the correlation between the instrument and the endogenous variable. The dependent variable in columns 2 is the Gini index, while in columns 3 and 4 the dependent variables are the average income of individuals located at the median point of the income distribution in USD and log respectively. All columns include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GPD of the finance sectors, and GDP of the government sectors. All variables are at Commuting Zone level and include Commuting Zone fixed-effects, year fixed-effects, and Commuting Zone-year trends. Errors are clustered at the Commuting Zone level: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Who Gains From Creative Destruction? Evidence From High-Quality Entrepreneurship in the United States

Appendix
(for online publication)

A-1 Robustness Checks for Shift-Share Instrument

To further validate the exogeneity of the shift-share instrument employed in the paper, we conduct a pre-trend analysis as suggested by Goldsmith-Pinkham et al. (2020). This involves examining if the residuals from a second-stage regression, where entrepreneurship is instrumented via the current instrument, are predictable by its future iterations. Table A5 shows the result of this exercise. The resulting coefficient is notably insignificant, reinforcing the instrument's validity.

Another recent methodology proposed by Adao et al. (2019) addresses the potential issue of the presence of residual correlation in shift-share regressions across commuting zones with similar industrial structures, regardless of geographical differences. This correlation can lead to standard inference methods over-rejecting the null hypothesis. The authors suggest two new inference methods that generate confidence intervals, considering the shift-share structure in residuals. Given that it is not possible to apply this method in the presence of multiple lags, we apply it to a simplified regression equation that does not include any lags while reporting the result of the corresponding standard instrumented regressions. The results in Table A6 show that confidence intervals calculated using Adao et al. (2019)'s methods align closely with those from standard IV procedures.

Additionally, Borusyak et al. (2020) propose a framework for testing the exogeneity of shocks, making sure that these are not correlated with specific features and characteristics at the area level. Table A7 presents the outcomes of several regressions where different features of commuting zone characteristics are regressed against the shocks. Assuming these shocks are randomly assigned, they should not significantly predict most of these variables. If they do, it suggests that locations more prone to substantial entrepreneurial shocks might exhibit specific features and experience unique dynamics. In general, results show that in our specification, shocks are unable to predict these controls which reassures us of their exogeneity.

A-2 Entrepreneurial Index (RECPI) and Other Entrepreneurial Measures

It is easier to appreciate the difference between the simple number of establishments as defined in the County Business Patterns (CBP) data and the entrepreneurial index (RECPI) (Andrews et al., 2019) by mapping the two measures for the states where the RECPI measure is available. Figure A3 maps the growth of both variables at the CZ level for the 1988–2014 period. Some areas correspond generally; in others, some divergent patterns can be observed. The number of establishments has grown substantially in states such as Idaho, Utah, Colorado, and Wyoming, while the RECPI index suggests that only a limited subset of CZs in these

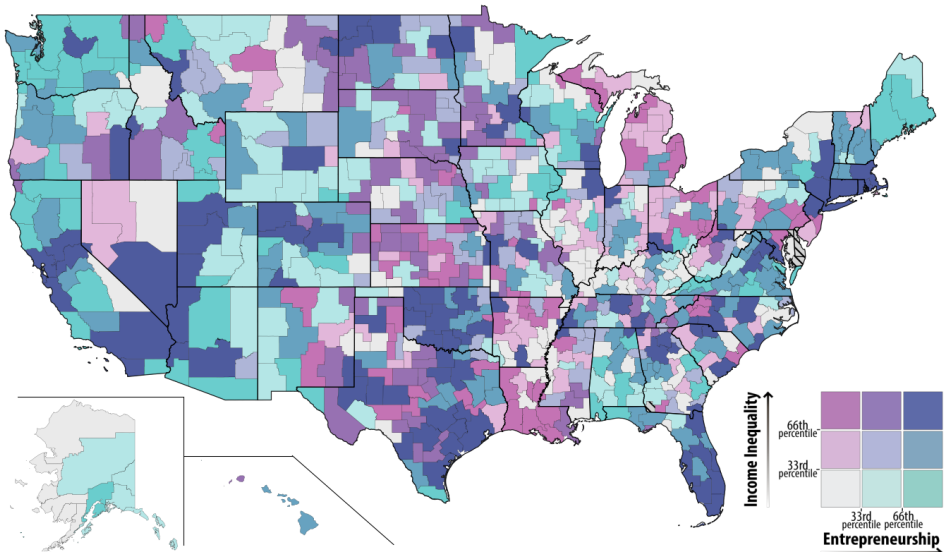
states have experienced significant growth in entrepreneurial quality. Another striking difference emerges when examining the two states with the best-known entrepreneurial hotbeds, California and Massachusetts. While the CZs belonging to these states have not experienced large growth in the number of firms, the growth of their entrepreneurial quality is remarkable.

To shed light on the validity of this measure of entrepreneurship, we plot the entrepreneurial index against total venture-capital investments (in dollars) in the United States, as well as the amount of investment by deal in the 1988–2014 period. Insofar as high-quality startups are more likely to attract external financing, given their higher potential to deliver substantial returns, these two measures should form a good proxy for assessing the presence of potentially high-growth firms. Figure A4 demonstrates that this variable tracks venture capital activity during the period. In particular, the RECPI index appears to be highly responsive to the dot-com bubble and the subsequent financial crisis, implying a strong link with the availability of financing opportunities for entrepreneurship.

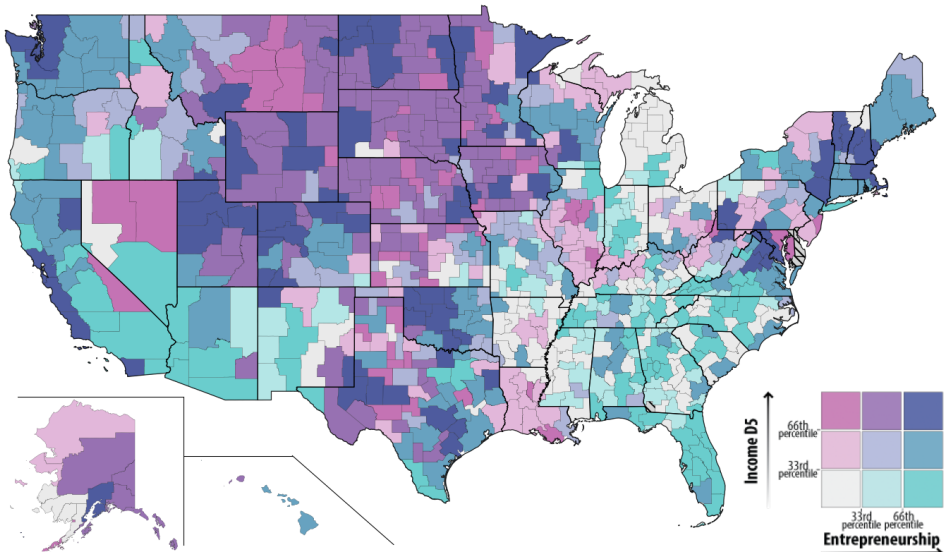
Appendix Figures and Tables

Figure A1: Bivariate Map of Entrepreneurship, the Gini Index, and the Average Income of the Median Individual (2014)

Panel A: Entrepreneurship and the Gini Index

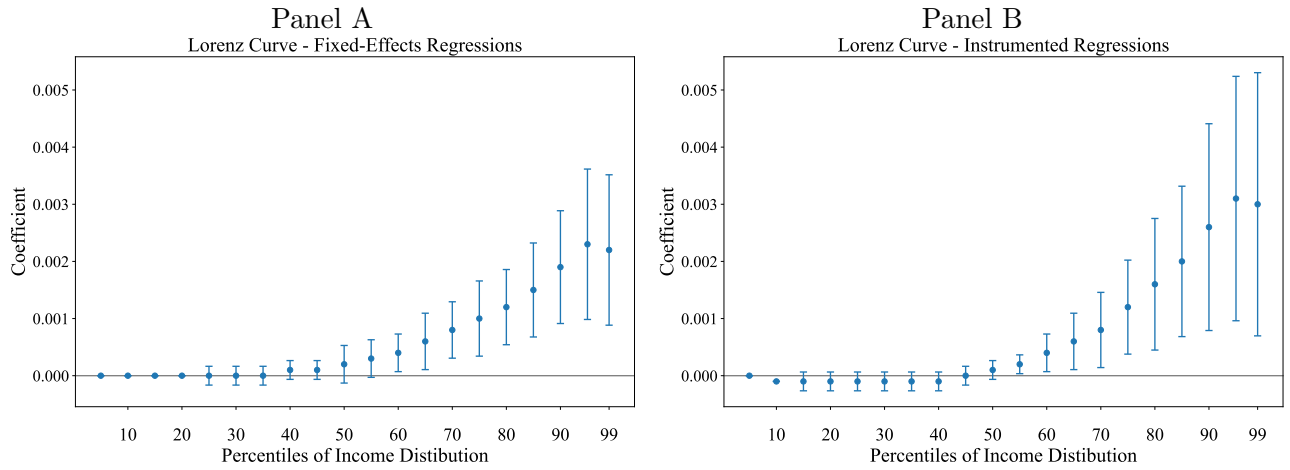


Panel B: Entrepreneurship and the Average Income of the Median Individual



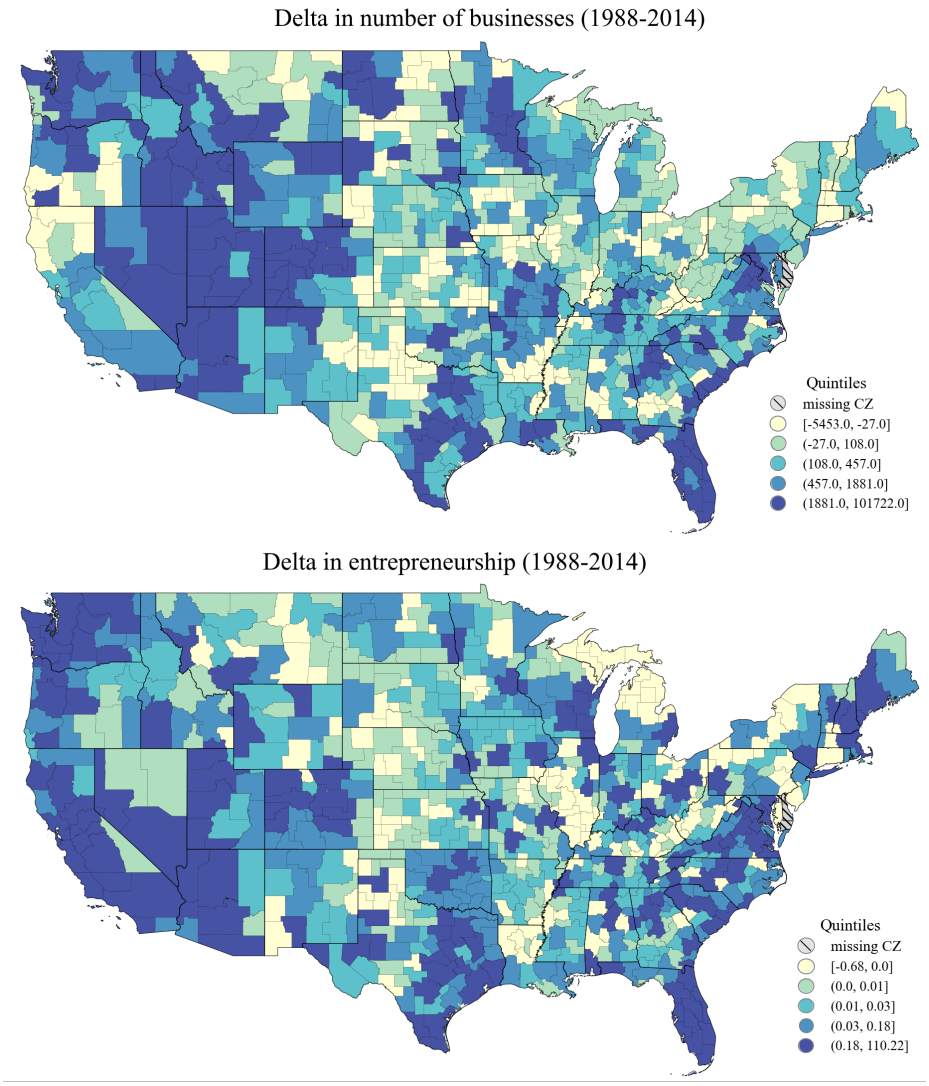
Sources: IRS Form 1040s, 1998–2014; Andrews et al. (2019).
Notes: The figure in Panel A shows the delta of the Gini index versus the delta of the RECPI index, while Panel B considers the average income of the median individual when considering the income distribution. Both figures consider the 1998–2014 period. Dark blue areas indicate CZs where the income measure (Gini or income) and RECPI index have increased considerably (both values exceed the 66th percentile of both distributions). White areas are CZs where the growth for both variables has not been particularly high (lower than the 33rd percentile in both distributions). Pink and cyan areas represent CZs where the considered income measure has increased considerably while the RECPI index has not, or vice versa. DRB approval numbers CBDRB-FY18-164, CBDRB-FY19-342, and CBDRB-FY19-509.

Figure A2: Coefficients of Fixed Effects and Instrumented Regressions of Lorenz Curve Percentiles on Entrepreneurship



Sources: IRS Form 1040s, 1998–2014; Andrews et al. (2019).
Notes: The graphs in Panel A present coefficients of several fixed-effects regressions (left figure) and instrumented regressions (right figure) where different Lorenz curve percentiles are regressed on entrepreneurship. All regressions include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GDP of the finance sectors, and GDP of the government sectors. Standard errors are clustered at the CZ level. The length of the vertical segments indicates 10-percent confidence-level intervals. IV regressions for which the F-statistic of the first stage is less than 10 are presented as dashed standard error lines. DRB approval numbers CBDRB-FY18-164, CBDRB-FY19-342, and CBDRB-FY19-509.

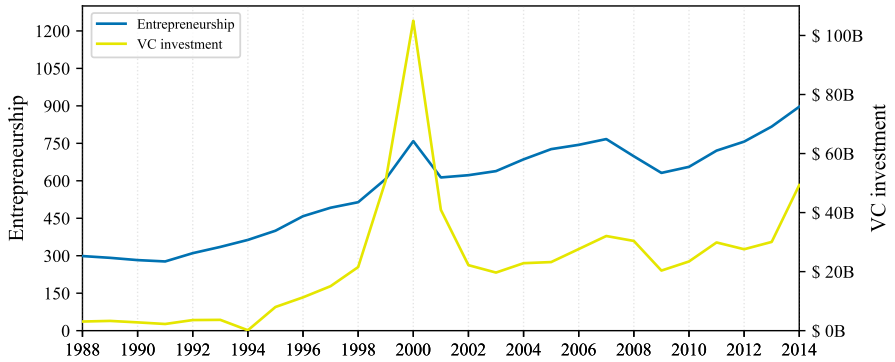
Figure A3: Growth (1988–2014) in the Log Number of Businesses Versus Growth in RECPI at the CZ Level (quintiles)



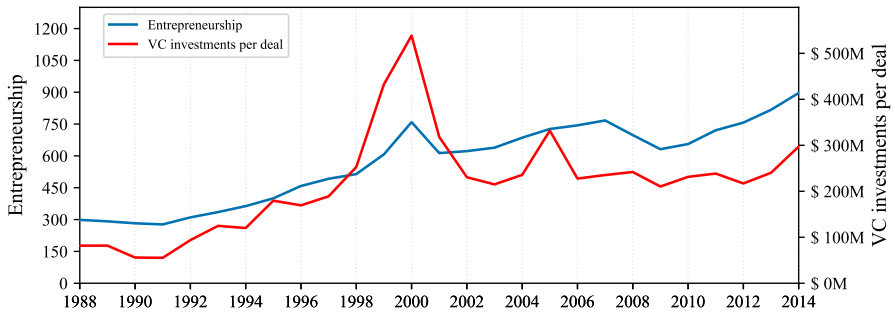
Sources: Andrews et al. (2019), County Business Patterns (CBP).
Notes: The figure shows the growth in the number of businesses (County Business Patterns) versus the growth of the RECPI index for the 1988–2014 period. Growth values have been calculated as the difference of the log values of the variables in 1988 and 2014. DRB approval numbers CBDRB-FY18-164, CBDRB-FY19-342, and CBDRB-FY19-509.

Figure A4: Entrepreneurship and Venture Capital

Panel A: Entrepreneurship (RECPI Index) and Venture Capital Investments From 1988 to 2014

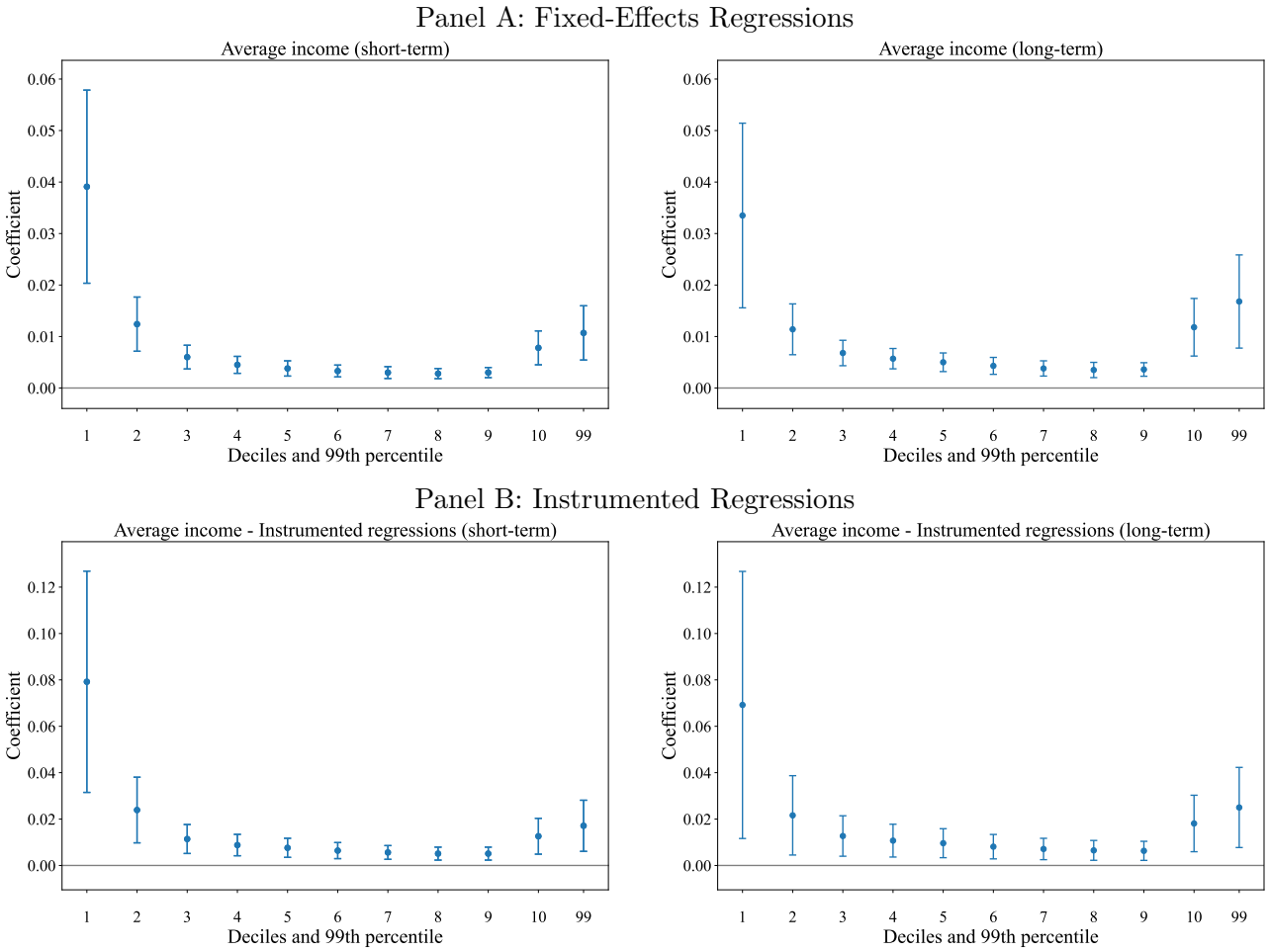


Panel B: Entrepreneurship (RECPI Index) and Investments by Deal From 1988 to 2014



Sources: National Venture Capital Association Yearbook 2015–2016, Andrews et al. (2019)

Figure A5: Coefficients of OLS and IV Regression of Logged Average Income by Deciles and 99th Percentile on Entrepreneurship



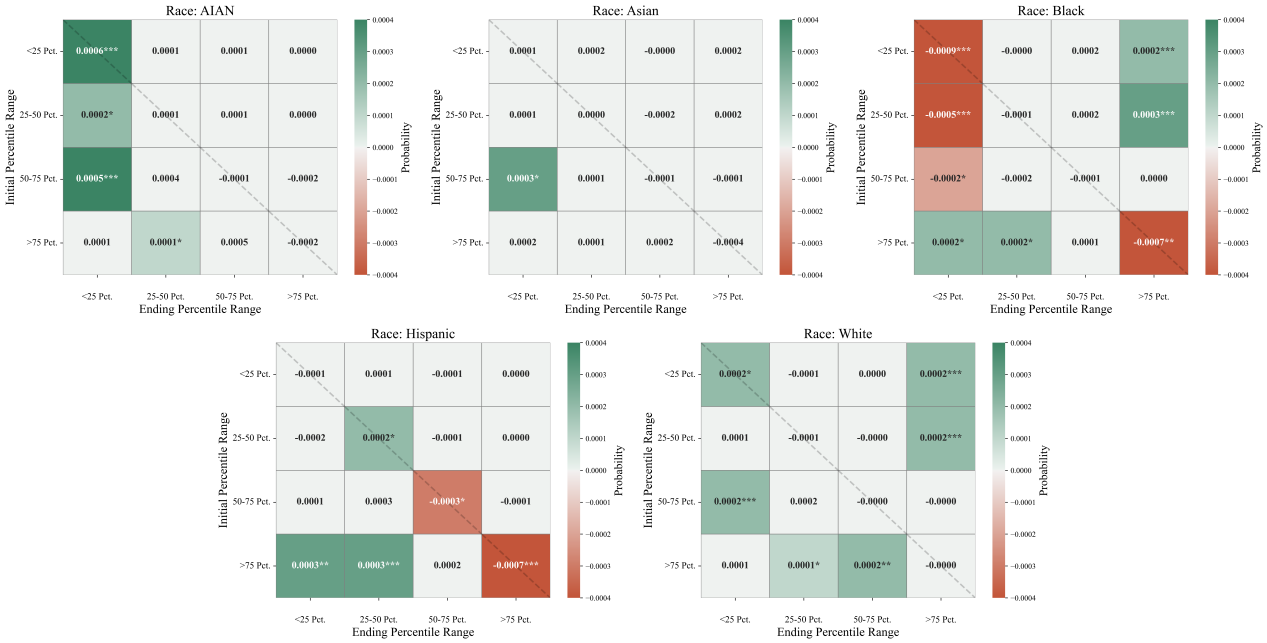
Sources: IRS Form 1040s, 1998–2014; Andrews et al. (2019). *Notes:* The graphs present coefficients of several fixed-effects (Panel A) and instrumented (Panel B) regressions where different levels of logged average income (which include all the deciles and the top 1 percent of the income distribution) are regressed on entrepreneurship. All regressions include controls for population, unemployment, race, age, GDP, female population, the share of employment of high-tech sectors, patents, GDP of the finance sectors, and GDP of the government sectors. Standard errors are clustered at the CZ level. The length of the vertical segments indicates 10-percent confidence-level intervals. IV regressions for which the F-statistic of the first stage is less than 10 are presented as dashed standard error lines. DRB approval numbers CBDRB-FY18-164, CBDRB-FY19-342, and CBDRB-FY19-509.

Figure A6: Coefficients of FE Regressions of the Probability of Belonging to a given Percentile Range by Initial Percentile Range after 5 years only Considering Non-Wage Incomes



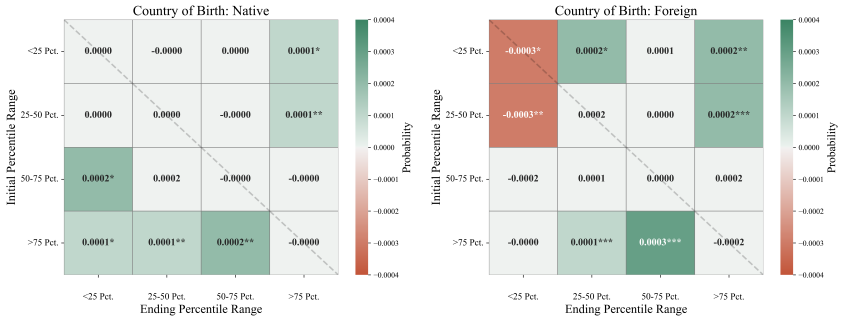
Sources: Kondo et al. (2023); Andrews et al. (2019). *Notes:* The graphs present coefficients of several fixed-effects regressions where measures of relative mobility (i.e., the probability of individuals belonging to the Xth percentile to belong to the Yth percentile after 5 years) are regressed on high-quality entrepreneurship. The graph considers only non-wage income. All regressions include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GDP of the finance sectors, GDP of the government sectors. Standard errors are clustered at the state level.

Figure A7: Coefficients of FE Regressions of the Probability of Belonging to a Given Percentile Range by Initial Percentile Range after 5 years by Race only Considering Non-Wage Incomes



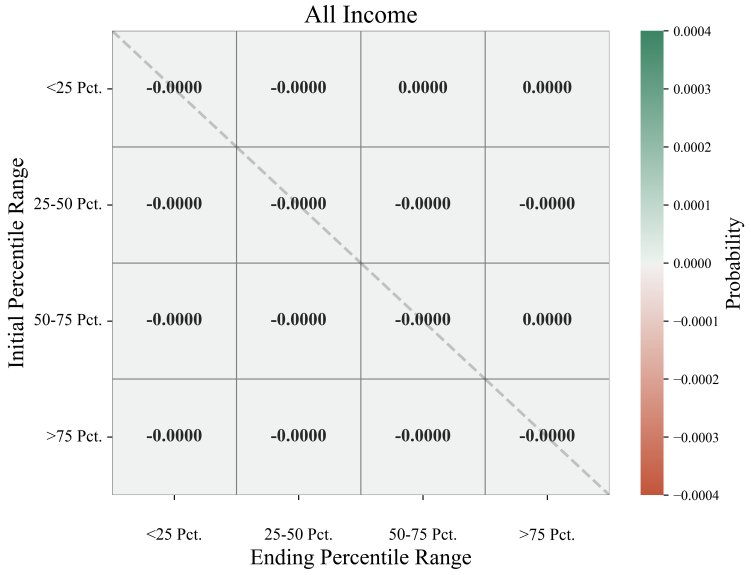
Notes: The graphs present coefficients of several fixed-effects regressions where mobility probabilities (measured as the probability of an individual belonging to a given percentile range to belong to another percentile range after 5 years) are regressed on the RECPI index considering only non-wage income. All regressions include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GDP of the finance sectors, GDP of the government sectors, and PCE. The regressions are run at the state-year level and include state- and year-fixed effects. Standard errors are clustered at the state level: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure A8: Coefficients of FE Regressions of the Probability of Belonging to a Given Percentile Range by Initial Percentile Range after 5 years by Country of Birth only Considering Non-Wage Incomes



Notes: The graphs present coefficients of several fixed-effects regressions where mobility probabilities (measured as the probability of an individual belonging to a given percentile range to belong to another percentile range after 5 years) are regressed on the RECPI index considering only non-wage income. All regressions include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GDP of the finance sectors, GDP of the government sectors, and PCE. The regressions are run at the state-year level and include state- and year-fixed effects. Standard errors are clustered at the state level: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure A9: Coefficients of FE Regressions of the Probability of Belonging to a given Percentile Range by Initial Percentile Range after 5 years (General Entrepreneurship)



Sources: Kondo et al. (2023); Andrews et al. (2019). *Notes:* The graphs present coefficients of several fixed-effects regressions where measures of relative mobility (i.e., the probability of individuals belonging to the Xth percentile to belong to the Yth percentile after 5 years) are regressed on a measure of entrepreneurship that takes into account the simple number of new companies. All regressions include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GDP of the finance sectors, GDP of the government sectors. Standard errors are clustered at the state level.

Table A1: The conceptual framework and its empirical counterparts

Mechanism (Section 2)	**Prediction**	**Empirical analysis**	**Finding**
Concentration of returns	Entrepreneurship widens the income distribution, concentrating gains at the top	Gini index and Lorenz / top income shares (Section 5)	Inequality rises, driven by larger income shares above the midpoint; the effect fades over time
Labor-market restructuring and diffusion	Entrepreneurship raises incomes across the whole distribution, not only at the top	Average income by decile (Section 5.2)	Incomes rise across the distribution in relative terms, most strongly at both tails, though absolute gains concentrate at the top; the left tail is lifted by lower-skill industries
Mobility dynamics	Creative destruction reshuffles positions, allowing upward mobility for some	Income transition matrices (Section 6)	Upward mobility from the bottom, offset by downside risk in the middle: a reordering rather than a compression
Structural barriers and demographic heterogeneity	Unequal access makes the benefits uneven across groups	Income and mobility by age, race, and nativity (Sections 5.2 and 6)	Gains are uneven across demographic groups, in both their size and the channel, wages or ownership, through which they flow

Table A2: Income ($000's) by demographic segment (P10, P50, P90)

	P10	P50	P90
By age			
Age 16–24	14.3	60.7	160.1
Age 25–34	22.1	70.1	157.2
Age 35–44	26.4	82.4	183.6
Age 45–54	28.1	90.4	207.4
Age 55–64	23.3	83.1	204.7
Age 65+	11.6	53.9	166.6
By race			
AIAN	14.8	60.2	147.8
Asian	22.2	84.6	215.3
Black	14.3	52.9	133.2
Hispanic	17.4	60.5	144.1
White	21.5	81.0	194.9
By status			
Foreign-born	19.4	71.2	182.5
Native	19.9	76.2	183.0
All	19.7	75.5	182.0

Sources: Kondo et al. (2023). *Notes:* Statistics at the state-year level for the period 1998–2014.

Table A3: Robustness: Fixed-Effects Regressions of Entrepreneurship on the Gini Index and Average Income of Individuals at the Median of the Income Distribution with State-by-Year and Division-by-Year Fixed Effects

	(1)	(2)	(3)	(4)
Model	FE	FE	FE	FE
Dep. Var.	Gini index	Gini index	Avg. Income (Log)	Avg. Income (Log)
Entrepreneurship	0.0004**	0.0005***	0.0041***	0.0044***
	(0.0002)	(0.0002)	(0.0010)	(0.0011)
CZ FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State-by-Year FE	Yes	No	Yes	No
Division-by-Year FE	No	Yes	No	Yes
N	11,644	11,695	11,644	11,695

Sources: IRS 1040s, 1998–2014; Andrews et al. (2019). *Notes:* Estimates of fixed-effects regressions. The dependent variable in columns 1 and 2 is the Gini index, while in columns 3 and 4 it is the logged average income of individuals located at the median point of the income distribution. All columns include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GDP of the finance sectors, and GDP of the government sectors. All variables are at the Commuting Zone level, and all columns include Commuting Zone and year fixed-effects. In place of the Commuting Zone-specific linear trends of the main specification, columns 1 and 3 add state-by-year fixed-effects, while columns 2 and 4 add Census division-by-year fixed-effects, so that the effect is identified from variation across Commuting Zones within the same state (or division) and year. Each Commuting Zone is assigned to a single state. The reported coefficient is the cumulative effect of entrepreneurship, summed across the contemporaneous and five lagged terms. Errors are clustered at the Commuting Zone level: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Robustess Checks of Fixed-Effects Regressions of Entrepreneurship on the Average Income of Individuals at the Median of the Income Distribution (Logged Values)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Model	FE	FE	FE	FE	FE	FE	FE	FE
Dep. Var.	Avg. Income 5th Decile (Log)	Avg. Income 5th Decile (Log)	Avg. Income 5th Decile (Log)	Avg. Income 5th Decile (Log)	Avg. Income 5th Decile (Log)	Avg. Income 5th Decile (Log)	Avg. Income 5th Decile (Log)	Avg. Income 5th Decile (Log)
Rob. Check	No Controls	Land Values	Average Rent	Large Est.	Exit Events	No MA	No CA	No MA & CA
Entrepreneurship	0.0052***	0.0051***	0.0037***	0.0045***	0.0052***	0.0053***	0.0105**	0.0236***
	(0.0016)	(0.0011)	(0.0010)	(0.0012)	(0.0011)	(0.0015)	(0.0051)	(0.0040)
N	12,002	11,695	9,555	11,695	11,695	11,611	11,423	11,339

Sources: IRS 1040s, 1998–2014; United States Department of Agriculture National Agricultural Statistics Service; Housing and Urban Development Health Economic and Residential Stability; County Business Patters; SDC Platinum; Andrews et al. (2019). *Notes:* Estimates of fixed-effects regressions with several robustness checks. The dependent variable in all columns is the average incomes of individuals located at the median point of the income distribution (logged values). Column 1 excludes all controls. Column 2 includes the value of land per acre (USD). Column 3 includes the median rent of a two-bedroom apartment (USD). Column 4 controls for the presence of large establishments. Column 5 controls for the number of exits (IPOs or significant mergers). Column 6 excludes CZs located in Massachusetts. Column 7 excludes CZs located in California. Column 8 excludes all CZ located in Massachusetts or California. All columns except Column 1 include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GPD of the finance sectors, and GDP of the government sectors. All variables are at the CZ level and include CZ, year fixed-effects, as well as CZ-year specific trends. Errors are clustered at the CZ level: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Pretrends Test (Goldsmith-Pinkham et al., 2020)

	(1)
Dependent Var:	Residuals
Instrument	0.001
	(0.001)
Observations	11,006

Sources: IRS 1040s, 1998–2014; Andrews et al. (2019). *Notes:* Results from pretrends tests (Goldsmith-Pinkham et al., 2017) using different dependent variables (the Gini index, the 90th Lorenz curve ordinate, and the average income of the top decile). All regressions include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GDP of the finance sectors, and GDP of the government sectors. All variables are at Commuting Zone level and include CZ fixed-effects, year fixed-effects and CZ-year trends. Errors are clustered at the CZ level:* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A6: AKM standard errors and confidence intervals (Adao et al., 2019)

	(1) Std.Error	(2) P-value	(3) 95% Conf. Interval
IV Standard inference	0.0014	0.037	0.0001, 0.0028
IV AKM	0.0014	0.000	0.0014, 0.0014
IV AKM0	0.0014	0.000	0.0014, 0.0014

The table presents standard errors, P-values and 95% confidence intervals of entrepreneurship instrumented with the shift-share instrument. These were obtained using the standard IV procedure and the two modified inference processes suggested by Adao et al. (2019).

Table A7: Balance Tests (Borusyak et al., 2020)

	(1) Coeff.	(2) Std. Err.
Perc. Pop. Asian	0.00001	(0.00001)
Perc. Pop. African-American	-0.00001	(0.00001)
Perc. Pop. Hispanic	0.00003	(0.00002)
Population	0.00003	(0.00011)
Perc. Pop. Over 65	-0.00001	(0.00002)
Perc. Pop. Under 25	-0.00001	(0.00002)
GDP	-0.00024	(0.00098)
Unemp. Rate	0.00001	(0.00003)
Female Pop.	0.003784	(0.09750)
Perc. Employed in High-Tech	-0.00001	(0.00001)
Patents	-0.20739	(0.58739)
GDP Finance Sector	-202.70894	(188.55985)
GDP Gov. Sector	12.76262	(131.18459)

The table reports results from OLS regressions where commuting zone features are regressed on shocks weighting by the average exposure shares. Robust standard errors are reported. (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$)

Table A8: Fixed-Effects Regressions of Entrepreneurship on the Gini index and Average Income of Individuals at the Median of the Income Distribution

	(1)	(2)
Model	FE	FE
Dep. Var.	Avg. Income 5th Decile (USD)	Avg. Income 5th Decile (Log)
Entrepreneurship	56.27*	0.0008*
	(29.91)	(0.0004)
N	600	600

Sources: IRS 1040s, 1998–2014; Andrews et al. (2019). *Notes:* Estimates of fixed-effects regressions. The dependent variables in Columns 1 and 2 are the average incomes of individuals located at the median point of the income distribution in USD and logged values, respectively. All columns include controls for population, unemployment, race, age, GDP, female population, share of employment of high-tech sectors, patents, GPD of the finance sectors, and GDP of the government sectors. All variables are at the state level and include state and year fixed-effects. Errors are clustered at the state level: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.